



UCL

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Machine Learning Analysis of Prefrontal fNIRS Data: Evaluating Autistic-Neurotypical Separability During Environmentally Complex Walking

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1. Abstract

Autistic adults may perceive complex visual environments differently, but it is unknown whether such differences translate strongly enough into prefrontal brain responses to support classification at the individual level. This dissertation tested whether functional near-infrared spectroscopy (fNIRS) measurements taken during walking under different lighting and obstacle conditions could separate autistic from neurotypical adults.

Condition-specific oxygenated-haemoglobin responses from 32 participants, 19 neurotypical and 13 autistic, were pooled into seven prefrontal regions. Four feature representations were assessed by three interpretable linear classifiers via repeated nested cross-validation and selection-aware permutation testing.

Descriptively, features describing the response between environmental conditions outperformed broad task-context averages. The strongest model, linear discriminant analysis using 28 contrast features, achieved a balanced accuracy of 0.619, high specificity to the neurotypical group (0.890), and low recall for the autistic group (0.348). This error was concentrated among a few autistic participants, consistent with within-group heterogeneity. Principal component analysis (PCA) reduced performance, while Sensory Perception Quotient (SPQ) scores added little predictive value. After accounting for selection across 12 model configurations, performance did not exceed random-label expectations ($p = 0.236$).

These findings are consistent with weak, context-dependent group structure, but do not establish it, and do not support a reliable or clinically useful classifier.

2. Introduction

Walking requires constant visual information interpretation and adaptation to changing environmental demands. Lighting, visual complexity, and obstacles may affect walking performance, cognitive effort, and the comfort with which one can navigate a route. Such demands may be particularly salient for autistic adults, who may perceive aspects of the visual and built environment differently than neurotypical adults. Because similar outward performance may mask differences in the effort or neural processing involved in maintaining it, the present dissertation asks whether prefrontal haemodynamic responses recorded during controlled walking conditions carry information that can discriminate autistic from neurotypical participants at an individual level.

2.1 Autism, Sensory Processing, and Built Environments

Autism is a neurodevelopmental condition characterised by persistent differences in social communication and interaction and restricted or repetitive patterns of behaviour, interests, or activities. Diagnostic criteria also include hyper- or hyporeactivity to sensory input, but sensory experiences vary widely across autistic individuals [1–3].

Features such as lighting, colour, sound, movement, visual clutter, and spatial layout can render a space overwhelming and difficult to navigate. In this regard, Black et al. [4]

outlined lighting and spatial organisation as issues of autism-inclusive design and emphasised adaptable lighting, simple layouts, and reduced sensory complexity. In this respect, Parmar et al. [3] reported that autistic adults described their sensitivity to light, motion, pattern, and certain colours as distracting, fatiguing, and stressful and that these sensitivities interfered with their ability to do everyday activities, such as travelling and using public spaces. Such experiences vary widely across individuals; thus, no environmental design can be presumed to accommodate the needs of all autistic people.

Thus, environmental design becomes relevant for autistic adults. However, perceiving an environment is not a sensory experience alone; it involves constant adaptation to the conditions of that environment.

2.2 Environmentally complex walking as a cognitive task

Whereas level walking depends, at least in part, on automatic motor processes, safe navigation in an environmentally complex setting requires attention, planning, and continuous adjustments. Walkers must monitor the route, integrate visual information, and adjust their movements in response to changes in environmental demands. Obstacles increase the need to plan foot placement and modify gait, whereas low illumination decreases visual information regarding the route and potential hazards.

The prefrontal cortex contributes to the attentional and executive control required when walking becomes more challenging. A recent systematic review of fNIRS studies suggested that increased complexity of walking commonly results in increased prefrontal cortical activation, especially when additional cognitive or motor control is involved in the tasks [5]. This aligns with increased use of attentional resources as environmental demands increase. Measuring prefrontal responses during actual walking may therefore reveal cognitive demands that are not evident from observable walking performance alone.

2.3 Measuring prefrontal responses during naturalistic walking

Functional magnetic resonance imaging (fMRI) allows excellent spatial resolution of brain activity but only if participants can remain still inside the scanner. Therefore, much of the research on the neural control of walking has employed imagined, simulated or constrained locomotion paradigms that may not capture the full spectrum of perceptual, motor and cognitive demands of real-life navigation [6,7].

Functional near infrared spectroscopy (fNIRS) is more suitable for studying brain responses during walking. Using near-infrared light, the apparatus estimates oxygenated and deoxygenated haemoglobin near the cortical surface through scalp-mounted optodes. Due to its portability and ability to be worn while standing and walking, fNIRS can measure prefrontal haemodynamic responses concomitant with movement in much more ecologically valid settings than traditional scanner-based imaging [5,8].

However, fNIRS remains vulnerable to measurement artefacts. Head movement can disturb optode to scalp contact, while heart rate, respiration and superficial blood flow can introduce signal changes unrelated to cortical activity. Careful signal-quality assessment, motion-artefact correction, filtering and preprocessing are therefore required before task-related haemodynamic responses can be interpreted [9].

2.4 Previous fNIRS evidence in autism

Previous fNIRS research suggests that autism may be associated with differences in both regional cortical activation and functional connectivity. In a systematic review of 30 studies, Zhang and Roeyers [10] identified recurring reports of atypical activation in the prefrontal cortex, inferior frontal gyrus and temporal cortices across tasks involving social processing, attention and working memory. Several studies also reported altered functional connectivity, indicating that autism-related differences may involve distributed neural organisation rather than isolated changes within a single region.

However, the location and direction of these effects varied across tasks, age groups and participant samples. This variability is consistent with the heterogeneity of autism and suggests that no single haemodynamic response is likely to characterise all autistic individuals. Relevant group information may instead be distributed across combinations of brain regions and experimental conditions, where it may not be apparent from separate group-average comparisons.

2.5 From group-average differences to individual-level classification

Conventional statistical analyses assess whether measurements differ between groups on average, but they do not establish whether those measurements can classify a previously unseen participant. Machine-learning classification addresses this separate question by learning a decision rule from labelled participants and evaluating it on held-out individuals. By combining features across brain regions and experimental conditions, machine learning may detect distributed information that is not represented by one consistently affected region [10,11].

Recent reviews indicate that diagnostic fNIRS machine-learning studies frequently combine modest sample sizes with flexible classifiers and high reported accuracies. Eken et al. [11] reviewed 45 clinical fNIRS machine-learning studies and found that SVM and deep-learning approaches were the most common, only eight studies enrolled more than 100 participants, and reported accuracy was negatively associated with sample size. This pattern suggests that high accuracy in small samples may not generalise, especially when preprocessing, model selection and evaluation are not fully separated. The present dissertation therefore did not seek to maximise apparent accuracy through more flexible classifiers. Instead, it prioritised simple interpretable linear models, participant-level nested cross-validation, training-only preprocessing, label-permutation testing and selection-aware maximum-statistic inference, with cautious interpretation rather than a diagnostic claim.

2.6 Existing PEARL walking-lighting dataset

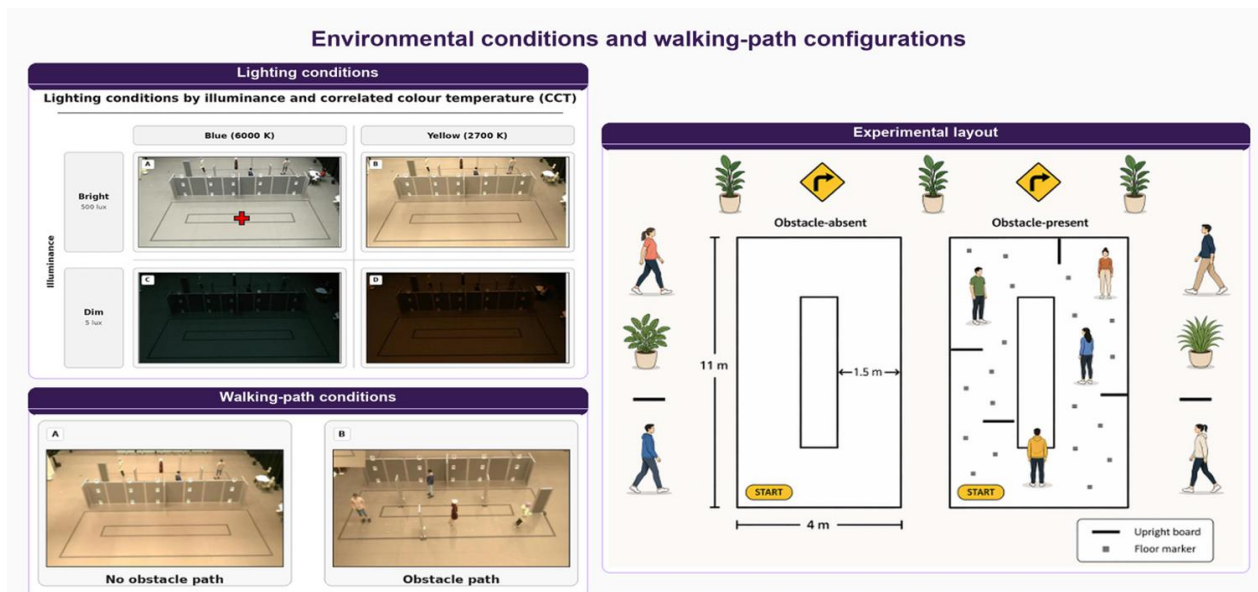
The source data were collected from 36 adults: 21 neurotypical and 15 autistic participants at UCL's Person-Environment-Activity Research Laboratory (PEARL), a full-scale controlled pedestrian environment designed to simulate real-world walking conditions. Participants completed eight walking trials comprising all combinations of illuminance, correlated colour temperature and obstacle presence [15].

Illuminance was varied between 5 and 500 lux, while correlated colour temperature was varied between 2700 K warm/yellow lighting and 6000 K blue-enriched lighting. Each lighting condition was completed with and without obstacles, producing eight conditions for every participant. Walking duration and prefrontal fNIRS activity were recorded during the task, while comfort and perceived physical and mental challenge were rated following each lighting condition. Participants also completed the Sensory Perception Quotient (SPQ) before the walking trials, which provided an exploratory measure of individual sensory differences. The experimental design is summarised in **Table 1**, with further details of the PEARL environment, route configuration, trial sequence and preprocessing details found in **Appendix A**.

Table 1. Summary of experimental factors.

Factor	Levels	Role in experiment
Illuminance	Bright: 500 lux; Dim: 5 lux	Manipulated lighting intensity
Correlated Colour Temperature	Blue-enriched: 6000 K; Yellow/warm: 2700 K	Manipulated spectral composition
Obstacle presence	Obstacle present; No obstacle	Manipulated environmental complexity

Figure 1. Environment setup in UCL PEARL. Adapted from Davoodian et al. [15].



2.7 Present study: aim and research questions

This study examined whether prefrontal fNIRS responses recorded during the PEARL walking task contained participant-level information separating autistic and neurotypical adults. It addressed three research questions:

1. Do prefrontal fNIRS features support autistic–neurotypical classification?
2. Are features representing relative responses between environmental conditions more informative than broad task-context averages?
3. Does the strongest observed classification performance exceed that expected from random group labels after accounting for selection across the evaluated model configurations?

Three secondary analyses further examined the structure and limitations of the observed classification results.

1. Are classification errors distributed across participants, or concentrated among particular autistic and neurotypical participants?
2. Are Sensory Perception Quotient (SPQ) scores associated with classification outcomes, and does adding SPQ improve classifier performance?
3. Can the full-contrast representation be compressed using PCA while retaining predictive performance and the original feature-level decision pattern?

Given the exploratory nature of the study, no directional hypotheses were specified. These questions were evaluated by comparing interpretable linear classifiers using repeated nested cross-validation and permutation testing.

3. Background

3.1 GLM/SPM beta estimation

Following preprocessing, task-related haemodynamic responses were estimated using a general linear model within the SPM framework. The GLM represents the measured fNIRS signal as a combination of experimentally defined predictors and residual error. In task-based fNIRS, condition timings are modelled as regressors and convolved with a haemodynamic response function to account for the delayed vascular response [9,16].

Ye et al. [16] introduced NIRS-SPM as a statistical parametric mapping framework in which fNIRS data can be analysed using the GLM to estimate haemodynamic responses across channels. The model can be written as:

$$Y = X\beta + \varepsilon \quad (1)$$

Where:

(Y) = Measured fNIRS haemodynamic signal over time,

(X) = Design matrix

(β) = Estimated beta coefficients

(ε) = Unexplained residual error.

The same experimental regressors were applied across channels; however, each channel had its own HbO time series and therefore its own set of beta estimates. Each beta coefficient represented the estimated response amplitude associated with a particular condition and channel.

These channel and condition-specific beta estimates formed the starting point for the present analysis. They were aggregated into predefined prefrontal regions of interest and converted into task-context and environmental contrast features. A fuller explanation of the design matrix, HRF convolution and beta estimation is provided in **Appendix B**.

3.2 Machine learning validation in neuroimaging datasets

Small neuroimaging datasets are vulnerable to optimistic classification estimates when model selection and performance evaluation are not adequately separated. Nested cross-validation reduces this bias by restricting model optimisation to the training data and evaluating the resulting procedure on held-out participants [12]. Performance estimates can nevertheless be uncertain when participant numbers are low and cross-validation folds overlap [13,14].

Permutation testing provides a nonparametric test of whether observed classification performance is better than expected if the group labels were independent of the features. An empirical null distribution can be generated by reassigning labels several

times and rerunning the entire modelling pipeline [17,18]. It is generally accepted that in neuroimaging, where classification often depends on cross-validation folds, more straightforward theoretical tests against chance are not valid [19,20]. If several prespecified model configurations are compared, retaining the best performance from each permutation also yields a selection-aware reference distribution for the best-observed result.

4. Methodology

This secondary analysis evaluated four label-independent fNIRS feature representations using three linear classifiers, producing 12 primary classifier–feature configurations. Balanced accuracy was the primary performance metric. PCA and SPQ analyses were treated as exploratory extensions, while selection across the 12 primary configurations was addressed using maximum-statistic label-permutation testing.

4.1 Participants and source data

The present study conducted a secondary analysis of data from 36 adults: 21 neurotypical and 15 autistic participants. Participants were recruited through university mailing lists and flyers distributed within the community. All participants in the autistic group reported a clinical autism diagnosis, although diagnoses were not independently verified.

Neurotypical participants had a mean age of 25.2 years (SD = 6.74; range 18-37 years), and autistic participants had a mean age of 29.5 years (SD = 7.42; range 20-40 years). The demographic characteristics are presented in Table 2.

Table 2. Participant demographic characteristics for the original 36-participant cohort.

Category	Characteristic	Neurotypical Adults (n = 21)	Autistic Adults (n = 15)	Total (n = 36)
Age	Age (Years)	M = 25.2 (SD = 6.74), Range = 18 – 37	M = 29.5 (SD = 7.42), Range = 20 – 40	-
Sex	Female, n (%)	5 (23.8%)	6 (40%)	11 (30.6%)
	Male, n (%)	16 (76.2%)	8 (53.3%)	24 (66.7%)
	Prefer not to say	-	1 (6.7%)	1 (2.8%)
Education	GCSEs / A-levels	10 (47.6%)	4 (26.7%)	14 (38.9%)
	Undergraduate / Postgraduate	10 (47.6%)	9 (60%)	19 (52.8%)
	Doctoral Degree	1 (4.8%)	0 (0%)	1 (2.8%)

	Other	0 (0%)	2 (13.3%)	2 (5.6%)
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Eligibility criteria included normal or corrected-to-normal vision, no history of a neurological condition associated with impaired navigation, and the ability to walk unaided for at least 20 minutes. All participants provided written informed consent. Ethical approval was granted by the UCL Research Ethics Committee (reference 5975/003).

4.2 SPM validation, beta extraction and ROI aggregation

The present analysis began from participant-level SPM.mat files generated during the original first-level fNIRS analysis. Prefrontal activity had been recorded using a portable Shimadzu LIGHTNIRS system, producing 22 measurement channels across the bilateral inferior frontal gyrus, dorsolateral prefrontal cortex, frontopolar cortex and medial prefrontal cortex [15]. The raw optical signals were not reprocessed and the GLM was not refitted; full acquisition and preprocessing details are provided in **Appendix A.4 and A.5**.

For each participant, the design matrix (SPM.xX.X), regressor names (SPM.xX.name) and stored beta estimates were extracted. Files were checked to confirm that these components were present, dimensionally compatible and could be matched by regressor name. Constant terms were excluded because they represented model intercepts rather than task-related responses. The required regressors represented the eight walking-lighting conditions, auditory events and transition periods, as summarised in Table 3.

Table 3. Interpretation of task/event regressor labels used in the SPM design matrix.

Regressor	Meaning
YDO	Yellow, dim, obstacle-present walking trial
YDN	Yellow, dim, no-obstacle walking trial
YBO	Yellow, bright, obstacle-present walking trial
YBN	Yellow, bright, no-obstacle walking trial
BDO	Blue, dim, obstacle-present walking trial
BDN	Blue, dim, no-obstacle walking trial
BBO	Blue, bright, obstacle-present walking trial
BBN	Blue, bright, no-obstacle walking trial
AE / AH	Auditory task/ event regressors
Transition	Transition periods between task blocks

Beta estimates were verified to be present and finite for all 22 channels in each retained participant; no channel contained missing, zero-valued or absent entries, and every

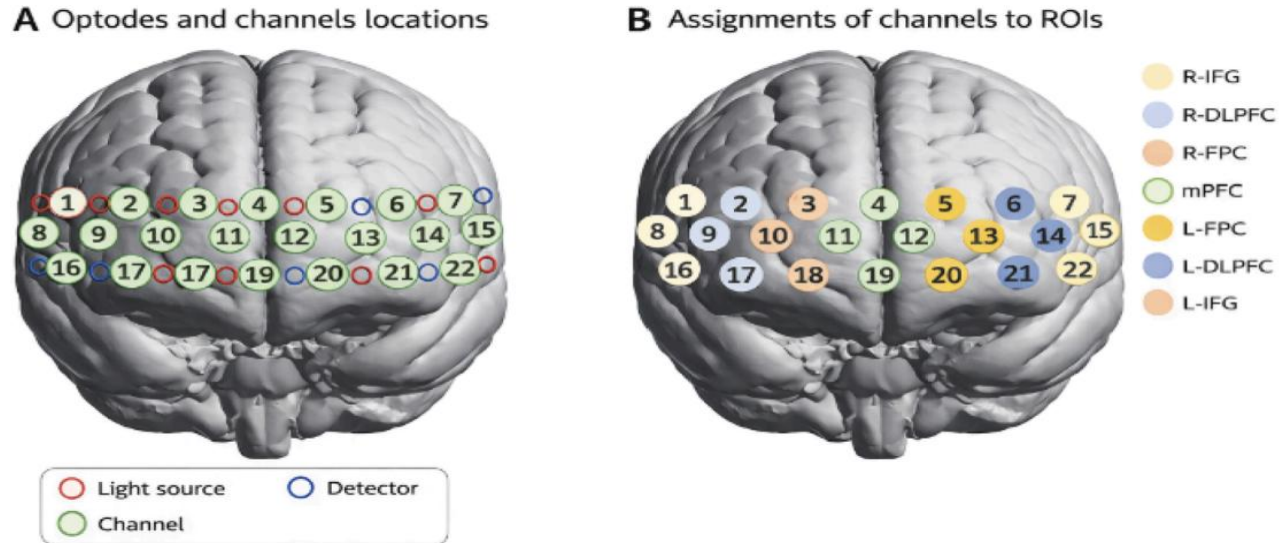
region-of-interest mean was therefore computed from its full channel complement. Channel-level quality decisions made during source preprocessing were not recorded in the supplied files and could not be re-applied.

For each retained participant, the condition-specific beta estimates were extracted across the 22 prefrontal channels and aggregated into seven predefined regions of interest. Channel values assigned to the same region of interest (ROI) were averaged, reducing dimensionality while retaining an anatomically interpretable representation. The channel-to-ROI assignments are summarised in **Table 4** and illustrated in **Figure 2**.

Table 4. Channel-to-ROI grouping.

Region of interest	Abbreviation	Assigned channels	Anatomical Interpretation
Right inferior frontal gyrus	R-IFG	1, 8, 16	Right lateral/inferior frontal region
Right dorsolateral prefrontal cortex	R-DLPFC	2, 9, 17	Right executive control region
Right frontopolar cortex	R-FPC	3, 10, 18	Right anterior prefrontal region
Medial prefrontal cortex	mPFC	4, 11, 12, 19	Medial anterior prefrontal region
Left frontopolar cortex	L-FPC	5, 13, 20	Left anterior prefrontal region
Left dorsolateral prefrontal cortex	L-DLPFC	6, 14, 21	Left executive-control region
Left inferior frontal gyrus	L-IFG	7, 15, 22	Left lateral/inferior frontal region

Figure 2. Optodes layout and region of interest (ROI) assignment across the prefrontal cortex. Adapted from Davoodian et al. [15].



The resulting condition-by-ROI beta values formed the basis of the feature-construction procedure described in Section 4.3.

4.3 Feature engineering

After SPM validation and ROI aggregation, each participant was represented using features derived from the condition-specific HbO beta estimates. Four label-independent feature representations were included in the primary analysis and were each evaluated with the three classifiers, producing the 12 prespecified classifier-feature configurations.

The task-context representation summarised broad haemodynamic responses during yellow-light walking, blue-light walking and non-walking/event periods. For each ROI, yellow walking was calculated by averaging YDO, YDN, YBO and YBN; blue walking by averaging BDO, BDN, BBO and BBN; and non-walking/events by averaging AE, AH and transition regressors. This produced three values for each of the seven ROIs, giving 21 features per participant.

The full-contrast representation described within-participant differences between experimental conditions. Four contrasts were calculated separately for each ROI: blue minus yellow, dim minus bright, obstacle minus no obstacle, and non-walking minus walking. This produced 28 features per participant. Because these features expressed differences between conditions rather than absolute response levels, they were intended to reduce the influence of general participant-level beta offsets.

Two summary feature representations were also derived from the condition contrasts. For each contrast, eight aggregate descriptors were calculated from the seven ROI values: the mean across all ROIs, the mean across left-hemisphere ROIs, the mean across right-hemisphere ROIs, the difference between the left and right means, the

bilateral mean for the frontopolar cortex, dorsolateral prefrontal cortex and inferior frontal gyrus, and a measure of the absolute global response magnitude. Applying these eight descriptors to each of the four contrasts produced 32 contrast-summary features per participant. A narrower blue minus yellow-summary representation applied the same eight descriptors only to the blue-minus-yellow contrast, producing eight features per participant.

Table 5. Feature representations included in the primary classifier comparison.

Feature representation	Construction	Features per participant	Information represented
Task-context ROI	3 task-context averages x 7 ROIs	21	Broad responses during yellow walking, blue walking and non-walking/event periods
Full-contrast ROI	4 condition contrasts x 7 ROIs	28	ROI-specific within-participant differences between conditions
Contrast summary	4 contrast maps x 8 summary measures	32	Global, hemispheric and ROI-family patterns across all contrasts
Blue-yellow summary	1 contrast map x 8 summary measures	8	Summary of the response to lighting spectrum alone

Two secondary variants were evaluated outside the primary 12-configuration comparison. First, the participant-level Sensory Perception Quotient score was added to the 28 full-contrast features, producing a 29-feature representation. Second, principal component analysis was evaluated as an optional dimensionality-reduction step fitted within the nested cross-validation pipeline.

4.4 Classifiers and nested cross-validation

Each feature table contained one row per participant. Participant identifiers were retained for tracking, while group labels were coded as 0 for neurotypical participants and 1 for autistic participants. All remaining numeric columns were used as predictors.

Three linear classifiers were evaluated: shrinkage linear discriminant analysis, L2-regularised logistic regression and a linear support vector machine, with further detail

provided in **Appendix C**. Regularised linear models were selected because the number of predictors was relatively large compared with the available training sample, while their linear decision functions remained interpretable.

Shrinkage regularisation was necessary because the pooled within-class covariance matrix cannot be estimated reliably when 28 predictors are fitted to approximately 24 training participants. In this regime the sample covariance matrix is singular and cannot be inverted, so standard linear discriminant analysis is undefined. Shrinkage addresses this by combining the sample covariance with a scaled identity matrix, producing a well-conditioned estimate at the cost of some bias. The mixing weight was either selected automatically using the Ledoit-Wolf procedure [26] or drawn from a fixed grid during inner cross-validation, and the selected values are reported in **Table 10**.

Performance was estimated using repeated nested stratified cross-validation. The outer loop used four folds repeated 25 times, producing 100 held-out evaluations. Within each repeat, every participant appeared in an outer test fold once, with approximately 24 participants used for model development and 8 held-out participants for testing.

For each outer split, hyperparameters were selected using three-fold stratified cross-validation applied only to the outer-training participants. The complete pipeline was tuned using balanced accuracy, refitted on the full outer-training set and then evaluated once on the untouched outer-test fold.

Standardisation, optional PCA and SPQ median imputation were included within the scikit-learn pipeline and fitted using training data only. This prevented preprocessing and model selection from being influenced by held-out participants. The same outer split schedule was used across all classifier-feature configurations.

Table 6. Summary of hyperparameter grids for each model.

Model	Hyperparameters searched
Linear Discriminant analysis	lsqr solver; shrinkage = auto, 0.1, 0.3, 0.5, 0.7 or 0.9
Logistic regression	L2 regularisation; liblinear solver; balanced class weights; $C = 0.01, 0.1, 1$ or 10
Linear support vector machine	Linear kernel; balanced class weights; $C = 0.01, 0.1, 1$ or 10; probability estimates enabled for ROC-AUC

Performance metrics were calculated separately for each outer test fold and summarised across the 100 evaluations using the mean.

4.5 Evaluation metrics

Performance was evaluated using complementary overall, class-specific and continuous-score metrics. Autistic participants were treated as the positive class and neurotypical participants as the negative class. The corresponding confusion-matrix terms are defined in Table 7.

Table 7. Summary of class-label definitions.

Term	Meaning in this dissertation
True Positive (TP)	Autistic participants correctly predicted
False Negative (FN)	Autistic participants incorrectly predicted as neurotypical
True Negative (TN)	Neurotypical participants correctly classified as neurotypical.
False Positive (FP)	Neurotypical participants incorrectly predicted as autistic

Balanced accuracy was the primary metric used for model tuning and comparison because the analytical cohort contained 19 neurotypical and 13 autistic participants. It gives equal weight to performance in both groups:

$$\text{Balanced accuracy} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (2)$$

Autistic recall, also known as sensitivity, measured the proportion of autistic participants correctly identified:

$$\text{Autistic recall} = \frac{TP}{TP + FN} \quad (3)$$

Neurotypical specificity measured the proportion of neurotypical participants correctly identified:

$$\text{Neurotypical specificity} = \frac{TN}{TN + FP} \quad (4)$$

These two measures were interpreted together because improved identification could occur at the expense of the other.

Autistic precision measured the proportion of participants classified as autistic who were genuinely members of the autistic group:

$$\text{Autistic precision} = \frac{TP}{TP + FP} \quad (5)$$

The F1 score for the autistic class combined autistic precision and recall. It therefore penalised both missed autistic participants and neurotypical participants incorrectly classified as autistic:

$$F1 = 2 \left(\frac{\text{Autistic precision} \times \text{Autistic recall}}{\text{Autistic precision} + \text{Autistic recall}} \right) \quad (6)$$

Receiver operating characteristic area under the curve (ROC-AUC) assessed how well continuous classifier scores ranked autistic participants above neurotypical participants across possible classification thresholds. The receiver operating characteristic curve plots the true-positive rate against the false-positive rate:

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN} \quad (7)$$

An ROC-AUC of 0.5 indicates no ranking ability, whereas a value of 1.0 indicates perfect separation. Continuous autistic-class probabilities or decision scores rather than hard class predictions were used to calculate ROC-AUC.

The Matthews correlation coefficient summarised agreement between predicted and true labels using all four confusion-matrix outcomes:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (8)$$

MCC ranges from -1 to +1, where +1 indicates perfect classification, 0 indicates no overall association between predictions and true labels, and -1 indicates complete disagreement.

Each metric was calculated separately for every held-out outer fold and then summarised across the 100 outer evaluations using the mean. Confusion-matrix counts were also pooled for descriptive visualisation.

4.6 Permutation-based statistical inference

Permutation testing was used to determine whether the observed classification performance was greater than could reasonably occur if the fNIRS features contained no information about whether a participant was autistic or neurotypical.

Group labels were shuffled at the participant level while preserving the original class distribution of 19 neurotypical and 13 autistic participants. Participant identifiers and feature values remained unchanged. For each permutation, the complete nested cross-validation procedure was repeated, including training-fold preprocessing, inner-loop hyperparameter selection, model refitting and outer-fold evaluation. The resulting balanced accuracies therefore represented the performance obtainable when the relationship between the features and group labels had been removed.

The primary analysis evaluated 12 configurations formed by combining four feature representations with three classifiers. Two complementary permutation results were calculated.

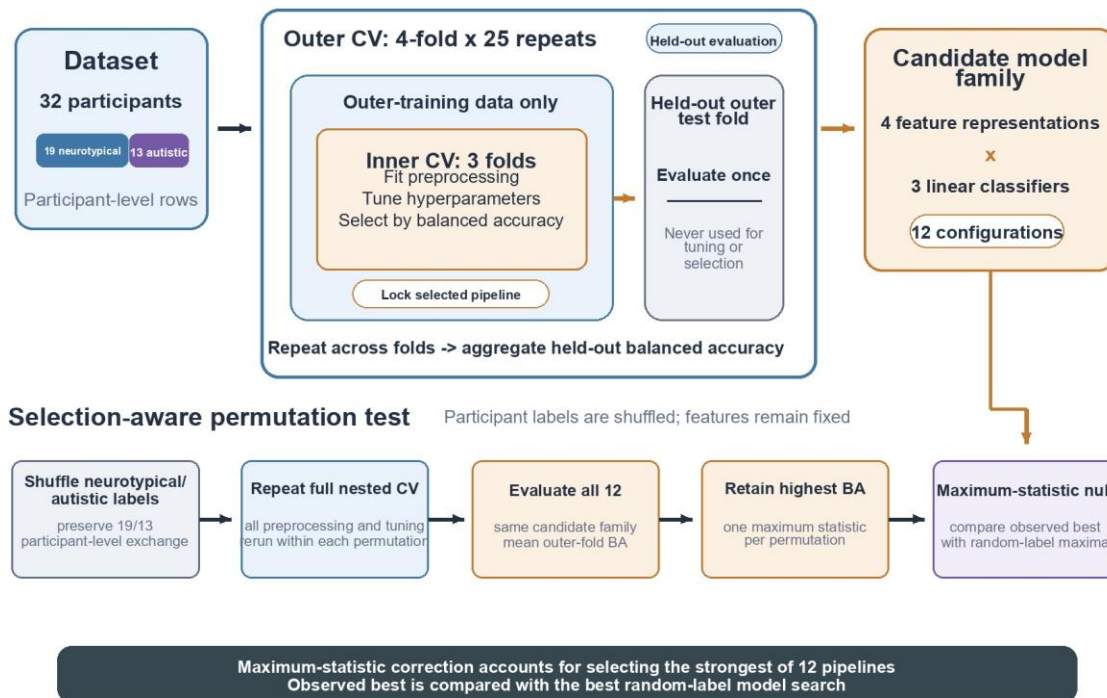
First, a configuration-specific p -value compared the observed balanced accuracy of each classifier-feature configuration with the permutation distribution generated for that same configuration. This assessed whether an individual configuration performed better than expected under random group labels when considered in isolation.

Second, a selection-adjusted maximum-statistic p -value accounted for comparison across all 12 configurations. For every permutation, the highest balanced accuracy obtained across the 12 configurations was retained. The strongest observed balanced accuracy was then compared with this distribution of best-performing random-label results. This provided a reference for how well the selected best model could perform simply because several configurations had been evaluated.

The maximum-statistic result was treated as the primary inferential test because the strongest observed configuration was identified after comparing all 12 configurations. A configuration-specific p -value does not account for this selection process and may therefore provide an overly favourable impression of the selected model when interpreted alone.

A total of 1,000 participant-label permutations were performed using a fixed random seed of 42. One-sided empirical p -values were calculated using a finite-sample correction, and statistical significance was assessed at $\alpha = 0.05$.

Figure 3. Leakage-controlled validation and selection-aware inference for small-sample neuroimaging classification.



4.7 Secondary and exploratory analyses

Following the primary comparison of the 12 classifier-feature configurations, additional analyses were conducted to characterise the strongest observed configuration and explore possible sources of variation in its predictions. The configuration with the highest mean outer-fold balanced accuracy was selected after completion of the primary model comparison. Its fitted coefficients and held-out participant predictions were summarised to examine the structure and stability of its decisions.

Separate exploratory analyses examined whether sensory-profile scores were associated with classification outcomes or fNIRS features, whether adding SPQ information changed classifier performance, and whether PCA improved performance through dimensionality reduction. These analyses were not included in the primary selection-adjusted permutation test and were interpreted as descriptive or exploratory rather than confirmatory, because SPQ introduced a supplementary questionnaire measure while PCA transformed the original anatomically interpretable feature space.

4.7.1 Exploratory coefficient analysis of the selected classifier

Coefficients were extracted from the model fitted within each of the 100 outer cross-validation splits. Because feature standardisation was performed separately within each split, the coefficients were interpreted within the corresponding standardised feature space. Positive coefficients shifted predictions towards the autistic class, whereas

negative coefficients shifted predictions towards the neurotypical class. Larger absolute coefficients indicated a greater contribution to the fitted decision function, although their magnitude could also be influenced by correlations between features.

For each feature, the median coefficient, interquartile range and 2.5th-97.5th percentile range were calculated across the outer fits. These summaries were used to describe the relative strength and consistency of the classifier's feature contributions. The analysis was exploratory and was not treated as a statistical test of regional group differences, biological importance or causal influence. Moreover, discriminative model weights do not necessarily correspond directly to neural activation patterns and should therefore be interpreted cautiously [22].

4.7.2 Participant-level prediction stability

Overall performance metrics describe average classifier behaviour but may conceal substantial variation between participants. Participant-level prediction stability was therefore examined for the selected configuration to determine whether errors were distributed broadly across the sample or concentrated among particular individuals, and whether classifications remained consistent when the composition of the training data changed.

Each participant appeared in an outer test fold once during each of the 25 repetitions, producing 25 held-out predictions made without that participant contributing to model training. For each participant, the proportion of correct predictions, proportion classified as autistic or neurotypical, majority-predicted class and distribution of decision scores were summarised. The number of errors contributed by each participant was also calculated to identify participants who were consistently classified correctly, repeatedly misclassified or sensitive to changes in the training sample.

These summaries were descriptive rather than inferential. Predictions from the same participant were repeated measures, and the corresponding outer-training sets overlapped across repetitions. Participant-level accuracy was therefore reported as the proportion of the 25 held-out predictions that matched the participant's true group label, without formal significance testing.

4.7.3 Statistical analyses for secondary evaluation

The secondary analyses used a rank-based group comparison, correlation coefficients and vector-similarity measures. The two-sided Mann-Whitney U statistic was calculated as:

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 \quad (9)$$

Where:

U_1 and U_2 = group-specific Mann–Whitney statistics;

n_1 and n_2 = subgroup sample sizes;

R_1 = rank sum for group 1.

Pearson's correlation coefficient measured linear association:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (10)$$

Where:

r = Pearson's correlation coefficient;

x_i and y_i = Paired values for participant i ;

$\bar{x}$ and $\bar{y}$ = Sample means of variables x and y , respectively;

n = Number of paired observations;

i = Index for each participant or observation.

Whereas, Spearman's rank correlation coefficient measured the strength and direction of a monotonic relationship between ranked observations:

$$\rho_s = \text{corr}(\text{rank}(x), \text{rank}(y)) \quad (11)$$

Both r and ρ_s range from -1 to $+1$; their signs indicate direction and their absolute magnitudes indicate association strength.

The Benjamini–Hochberg false-discovery-rate procedure was applied as follows. After ordering the mp -values such that $p_{(1)} \leq \dots \leq p_{(m)}$, the largest rank k satisfying (12) was identified.

$$p_{(k)} \leq \frac{k}{m} Q \quad (12)$$

Where:

$p_{(k)}$ = the k -th ordered p -value;

k = rank of the ordered p -value;

m = total number of tests;

Q = prespecified false-discovery-rate level.

All hypotheses corresponding to $p_{(1)}, \dots, p_{(k)}$ were rejected and adjusted q -values were reported.

Cosine similarity quantified the directional alignment of two coefficient vectors:

$$\cos(\theta) = \frac{w_{PCA} \cdot w_{non-PCA}}{\|w_{PCA}\| \|w_{non-PCA}\|} \quad (13)$$

A value approaching +1 indicated closely aligned decision directions, whereas values near zero indicated weak alignment.

4.7.4 Exploratory SPQ analysis

The Sensory Perception Quotient (SPQ) is a participant-level self-report measure of sensory perception across vision, hearing, touch, taste and smell, with lower scores indicating greater sensory sensitivity [21]. SPQ scores were matched to the fNIRS feature data using participant identifiers.

Three exploratory analyses were conducted. First, autistic participants were divided according to the held-out predictions produced by the selected classifier. Participants classified as autistic in at least 50% of their repeated predictions were assigned to the more frequently detected subgroup, whereas those classified as neurotypical in more than 50% were assigned to the more frequently missed subgroup. Given the small subgroup sizes and the absence of an assumed normal score distribution, SPQ scores were compared using a two-sided Mann-Whitney U test. The test assessed whether the rank distributions differed between the subgroups without specifying the direction of the difference in advance.

Second, associations between SPQ scores and the 28 contrast-ROI features were examined using Pearson and Spearman correlations. Pearson correlation assessed linear relationships between the original values, whereas Spearman correlation assessed monotonic relationships using ranked values and was less dependent on distributional assumptions. Benjamini-Hochberg false-discovery-rate correction was applied across the 28 Spearman correlation p-values, and adjusted q-values were reported. Pearson correlations were included as supplementary, uncorrected estimates of linear association.

Third, SPQ was added as a supplementary predictor to the full-contrast feature representation to determine whether sensory-profile information changed classifier performance. Missing SPQ values were imputed using the training-fold median. Performance with and without SPQ was compared descriptively across the three classifiers.

4.7.5 Exploratory PCA compression and interpretability analysis

Principal component analysis (PCA) was used as an exploratory sensitivity analysis to determine whether the 28 full-contrast features could be represented in fewer dimensions while retaining classification performance and the original feature-level decision pattern.

Following the primary model comparison, coefficient preservation was examined for the classifier taken forward for detailed interpretation. The matched PCA and non-PCA LDA

pipelines used identical cross-validation partitions, random seeds, scaling procedures and LDA shrinkage candidates. Both used the lsqr solver, while the PCA pipeline additionally tuned the number of retained components from one to 12.

Within each outer-training fold, StandardScaler and PCA were fitted using training participants only. Component counts from one to 12 were evaluated jointly with the classifier hyperparameters during three-fold inner cross-validation, using balanced accuracy for selection.

Because PCA identifies directions of overall variance without using group labels, component loadings were not treated as measures of classification importance. Instead, coefficients estimated in principal-component space were projected back into the original standardised feature space:

$$w_{back} = P^T w_{PC}, \quad (14)$$

Where:

(P) = Fitted PCA component matrix

w_{PC} = Classifier coefficient vector in component space

Coefficient orientation was aligned to the autistic class, and the reconstructed coefficients and intercept were numerically validated against the fitted pipeline decision scores.

Across matched outer fits, PCA and non-PCA coefficient patterns were compared using cosine similarity, Pearson and Spearman correlations, sign agreement, top-feature overlap, relative magnitude and rank changes. The proportion of the original decision direction retained within the PCA subspace was also calculated. All cross-fit summaries were reported as medians and interquartile ranges and interpreted descriptively, because the repeated models reused the same participants.

4.8 Software testing and pipeline validation

Automated tests used pytest to verify data processing, feature engineering, machine-learning input validation, participant-level prediction summaries, questionnaire processing and permutation behaviour. The final repository-root run passed 87 tests across seven modules, with two expected questionnaire-merge warnings. Software versions, random seeds and key analysis scripts are documented in **Appendix D**.

Table 8. Summary of validation performed for the machine-learning pipeline.

Pipeline area	Validation performed
`SPM.mat` validation	Checked matrix dimensions, 22-channel structure, regressor consistency, missing conditions and zero-valued task columns

ROI aggregation	Compared seven ROI means against manually calculated synthetic values
Feature engineering	Verified task-context averages and the direction and magnitude of the main contrast features
Feature-table loading	Rejected missing columns, invalid labels, non-numeric predictors and non-finite values
Model pipelines	Confirmed that imputation, scaling, optional PCA and classification were contained within scikit-learn pipelines
Nested cross-validation	Smoke-tested inner-loop tuning, outerfold evaluation, predictions and output generation
Prediction analysis	Tested participant-level accuracy, autistic prediction percentages and stability classifications
SPQ processing	Tested score conversion, duplicate identifiers, participant matching, warning and imputation placement
Pipeline area	Validation performed
Statistical analysis	Verified label shuffling, p-values, resumption and maximum-statistic correction

5. Results

5.1 Participant inclusion

The initial dataset contained 36 participants: 21 neurotypical (NT) and 15 autistic participants. A loadable SPM.mat file was available for all 36 participants, so no exclusions were attributable to missing or unreadable files.

Four participants were subsequently excluded during design-matrix validation: P53 and P59 from the neurotypical group, and P34 and P60 from the autistic group. In each case, at least one required task-regressor column contained only zero values, preventing generation of a complete and comparable feature set. The final analytical cohort therefore contained 32 participants: 19 NT and 13 autistic.

Table 9. Participant inclusion following SPM validation.

Processing stage	NT, n	Autistic, n	Total, n
Initial participant cohort	21	15	36
Participants with loadable SPM.mat files	21	15	36
Excluded following design-matrix validation	2	2	4
Final cohort	19	13	32

The supplied participant-level metadata verified the final group counts but did not contain age, sex or education fields; post-exclusion demographic summaries could therefore not be recalculated without an additional source file.

5.2 Core model comparison

Figure 4 summarises performance across the 12 combinations of four feature representations and three linear classifiers. The clearest descriptive pattern concerned feature representation rather than classifier choice. Across all three classifiers, the full-contrast and contrast-summary representations produced higher balanced accuracy and Matthews correlation coefficients than the broader task-context representation. In comparison, restricting the summary features to the blue-yellow contrast produced smaller improvements. Selected hyperparameters across the outer fits are listed in **Table 10**.

Table 10. Selected hyperparameters across the 100 outer fits

Configuration	Hyperparameter	Modal selected value	Selection frequency
Task_context_lda	Shrinkage	0.1	27/100
Task_context_logreg	C	0.01	41/100
Task_context_linear_svm	C	0.01	44/100
Contrast_lda	Shrinkage	Auto	30/100
Contrast_logreg	C	0.1	35/100
Contrast_linear_svm	C	0.01	35/100

Contrast_summary_lda	Shrinkage	0.9	26/100
Contrast_summary_logreg	C	0.01	40/100
Contrast_summary_linear_svm	C	0.01	36/100
Blue_yellow_summary_lda	Shrinkage	Auto	41/100
Blue_yellow_summary_logreg	C	0.01	46/100
Blue_yellow_summary_linear_svm	C	0.01	45/100

Figure 4. Summary of primary metrics across classifiers and feature sets.



Balanced accuracy for the two broader contrast representations ranged from 0.571 to 0.619, compared with 0.483 to 0.517 for the task-context representation. The highest mean balanced accuracy was obtained by LDA using the full 28-feature contrast representation at 0.619. The next-highest values were both 0.608, obtained by contrast-summary LDA and contrast-summary logistic regression. The corresponding MCC values were also highest for the contrast-based configurations, with full-contrast LDA producing the largest value at 0.286.

A similar pattern was observed for ROC-AUC with LDA and logistic regression. Their full-contrast and contrast-summary configurations produced ROC-AUC values between

0.675 and 0.697, whereas their task-context values were below 0.50. Linear SVM showed a different pattern: its contrast-based balanced accuracies were above 0.57, but its corresponding ROC-AUC values remained below 0.50. Thus, its thresholded classifications and continuous score rankings did not show the same ordering.

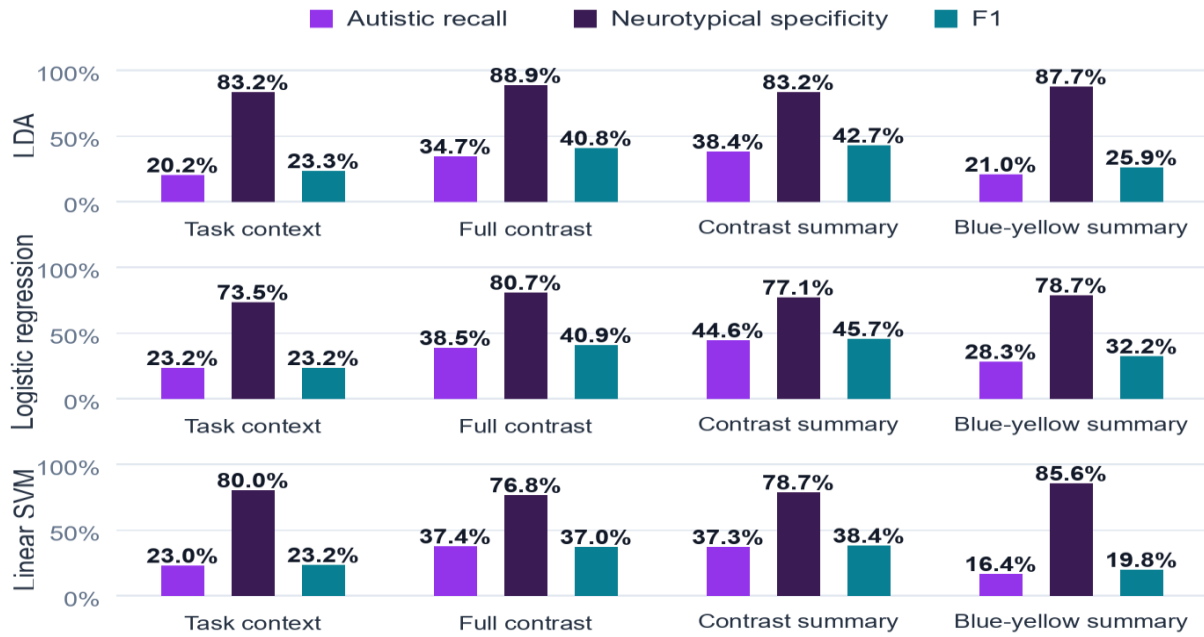
This dissociation is consistent with the way continuous scores were obtained for the linear SVM. Unlike LDA and logistic regression, the SVM does not produce class probabilities directly, and probability estimates were generated by a sigmoid calibration fitted through an internal cross-validation step within each training fold (Appendix C.3). With approximately 24 training participants, this calibration was estimated from very limited data. Because the probability model was fitted separately using internal cross-validation, the calibrated probabilities could be inconsistent with the hard class predictions produced by the fitted SVM. The SVM ROC-AUC values should therefore be interpreted cautiously and were not used for model selection, which was based on balanced accuracy throughout.

Overall, representations describing relative responses across multiple experimental conditions produced stronger descriptive performance than broad task-context averages or summaries restricted to colour temperature alone. Full-contrast LDA was designated the strongest observed configuration because it achieved the highest mean outer-fold balanced accuracy. It was therefore selected for the subsequent exploratory coefficient and participant-stability analyses. Whether this performance exceeded that expected under random group labels is evaluated separately in **Section 5.4**.

5.3 Confusion matrices and class-specific performance

Class-specific performance is summarised in **Figures 5** and **6**. The confusion matrices pool predictions across 100 repeated outer-fold evaluations, producing 800 held-out predictions for each classifier-feature configuration. Because each participant contributed 25 predictions, these counts represent repeated held-out classifications rather than 800 independent participants.

Figure 5. Autistic recall, neurotypical specificity and F1 score across classifier-feature configurations. (Calculated separately across the 100 outer-fold evaluations)



The clearest pattern was a consistent imbalance between the two groups. Across all 12 configurations, neurotypical specificity exceeded autistic recall. Autistic recall ranged from 0.164 to 0.446, whereas neurotypical specificity ranged from 0.735 to 0.889. The dominant error was therefore the classification of autistic participants as neurotypical, and no configuration identified at least half of the autistic participants on average.

A general sensitivity-specificity trade-off was also evident across classifiers. LDA typically favoured neurotypical specificity, whereas logistic regression generally increased autistic recall at the cost of more neurotypical participants being classified as autistic. For example, with the contrast-summary representation, logistic regression produced the highest autistic recall across all configurations (0.446), but lower neurotypical specificity (0.771) than LDA (0.832).

Figure 6. Confusion-matrix summary across classifiers and feature sets. (Percentages here were calculated from pooled counts across the 800 repeated held-out predictions for each configuration.) Here neurodivergent (ND) refers to autistic participants

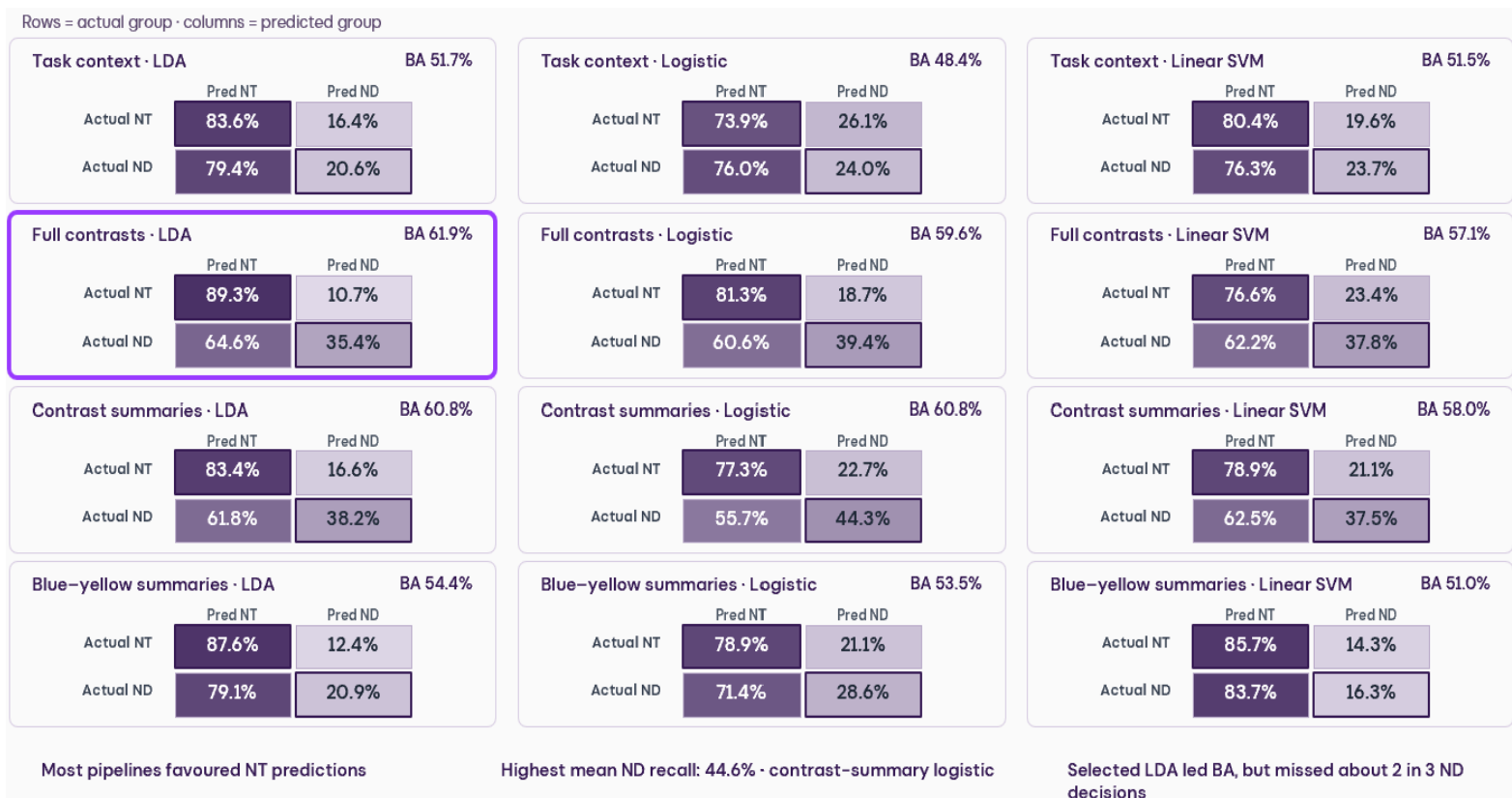


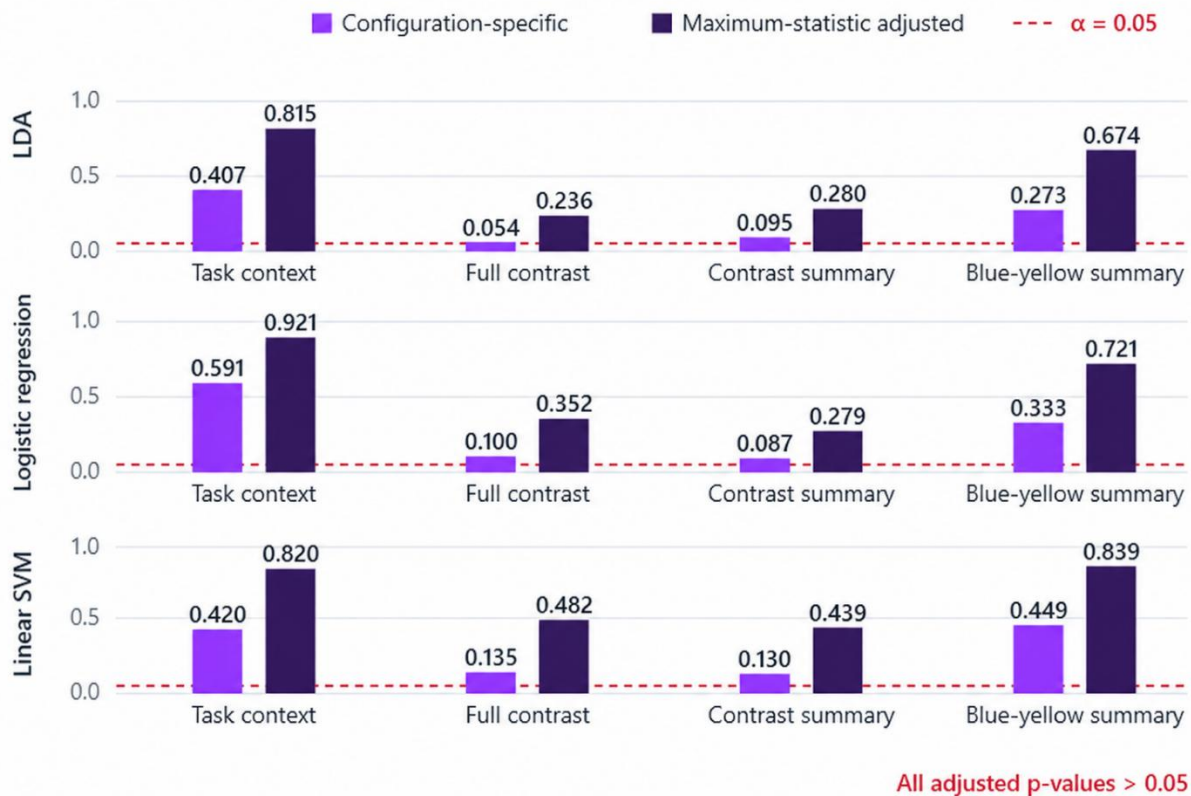
Figure 6 further illustrates this imbalance: full-contrast LDA produced a false-negative rate of 64.6% among autistic participants, compared with a false-positive rate of 10.7% among neurotypical participants. Its comparatively strong balanced accuracy was therefore driven more by reliable neurotypical classification than by balanced identification of both groups.

Overall, the classifiers differed mainly in how they distributed errors between autistic and neurotypical participants. None of the evaluated configurations achieved balanced class-specific performance.

5.4 Permutation-based statistical inference

Permutation testing was used to determine whether the strongest observed classification performance exceeded that expected when group labels were unrelated to the fNIRS features. All 1,000 participant-label permutations were completed successfully, with each permutation repeating the full nested cross-validation procedure across the same 12 classifier-feature configurations.

Figure 7. Configuration-specific and selection-adjusted permutation p-value across 1,000 permutation runs.



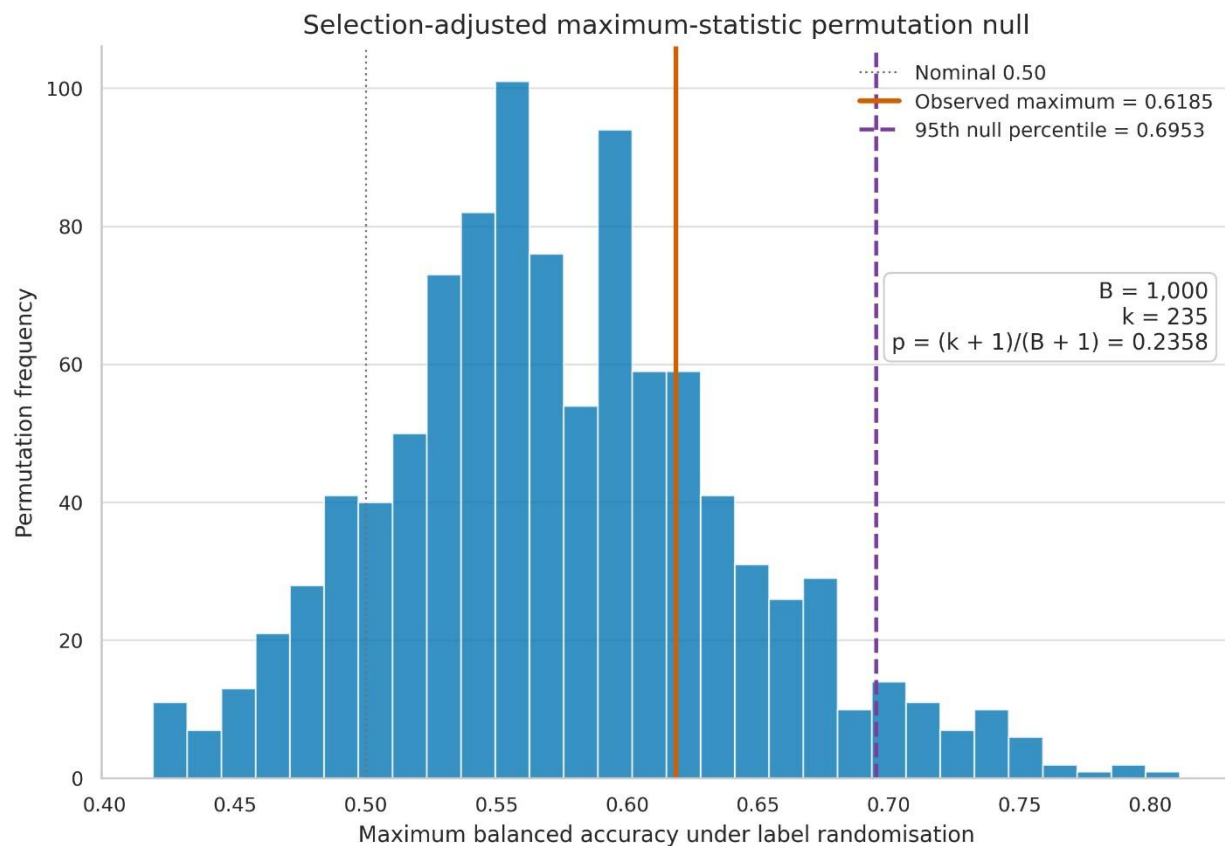
Configuration-specific and selection-adjusted results are summarised in **Figures 7** and **8**. Full-contrast LDA produced the highest observed balanced accuracy of 0.6185 and the smallest configuration-specific p -value of 0.0539. This result did not meet the prespecified significance threshold of $\alpha = 0.05$. Therefore, even when considered in isolation, full-contrast LDA did not provide statistically significant evidence of performance above its random-label null distribution.

However, full-contrast LDA was identified only after comparing all 12 configurations. The selection-aware maximum-statistic test was therefore treated as the primary inference. For each permutation, the highest balanced accuracy across the 12 configurations was retained, providing a null distribution that reflected the performance obtainable after searching across multiple models.

The maximum balanced accuracy under label permutation had a mean of 0.5744, compared with the observed maximum of 0.6185. The observed result therefore exceeded the average random-label maximum by approximately 0.044. Nevertheless, 235 of the 1,000 permutations produced a maximum balanced accuracy equal to or greater than the observed value, yielding a finite-sample-corrected selection-adjusted p -value of 0.2358.

The null hypothesis was therefore not rejected. After accounting for selection across the 12 evaluated configurations, the study did not establish that the strongest observed classifier performed better than could reasonably occur under random group labels. Full-contrast LDA can consequently be described as the strongest observed configuration in this sample, but not as evidence of statistically reliable autistic-neurotypical classification.

Figure 8. Observed maximum against the permutation distribution.

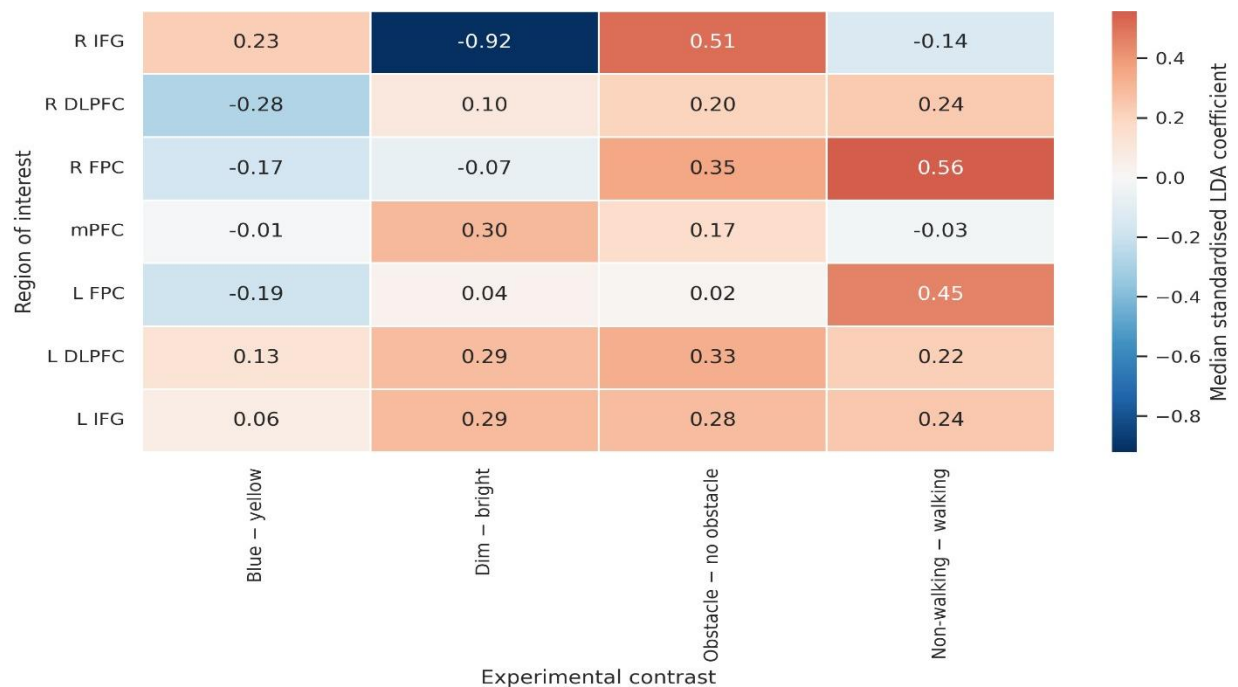


5.5 Full-contrast LDA coefficient structure

Coefficients from the 100 outer-training full-contrast LDA models were examined to describe how the selected classifier combined the 28 contrast-ROI features. Positive coefficients indicated that larger feature values shifted the decision score towards the autistic class, whereas negative coefficients shifted it towards the neurotypical class.

The principal pattern was distributed and condition-dependent. No single prefrontal region contributed in one consistent direction across all experimental contrasts. Instead, the contribution of a region depended on the environmental comparison being represented. This was most apparent in the right inferior frontal gyrus: its median coefficient was positive for blue minus yellow (+0.23) and obstacle minus no obstacle (+0.51), but negative for dim minus bright (−0.92) and non-walking minus walking (−0.14). The same region could therefore shift the classifier towards either group depending on the contrast.

Figure 9. Median fold-fitted LDA coefficients by contrast and ROI.



The pattern also differed across contrasts. Obstacle minus no obstacle coefficients were positive across all seven ROIs, although their magnitudes varied. Non-walking minus walking contributions were strongest in the right and left frontopolar cortices (+0.56 and +0.45), whereas blue minus yellow coefficients were generally smaller and mixed in direction. The classifier therefore appeared to combine several regional and condition-specific contributions rather than rely on a single putative “autism region” or one environmental manipulation.

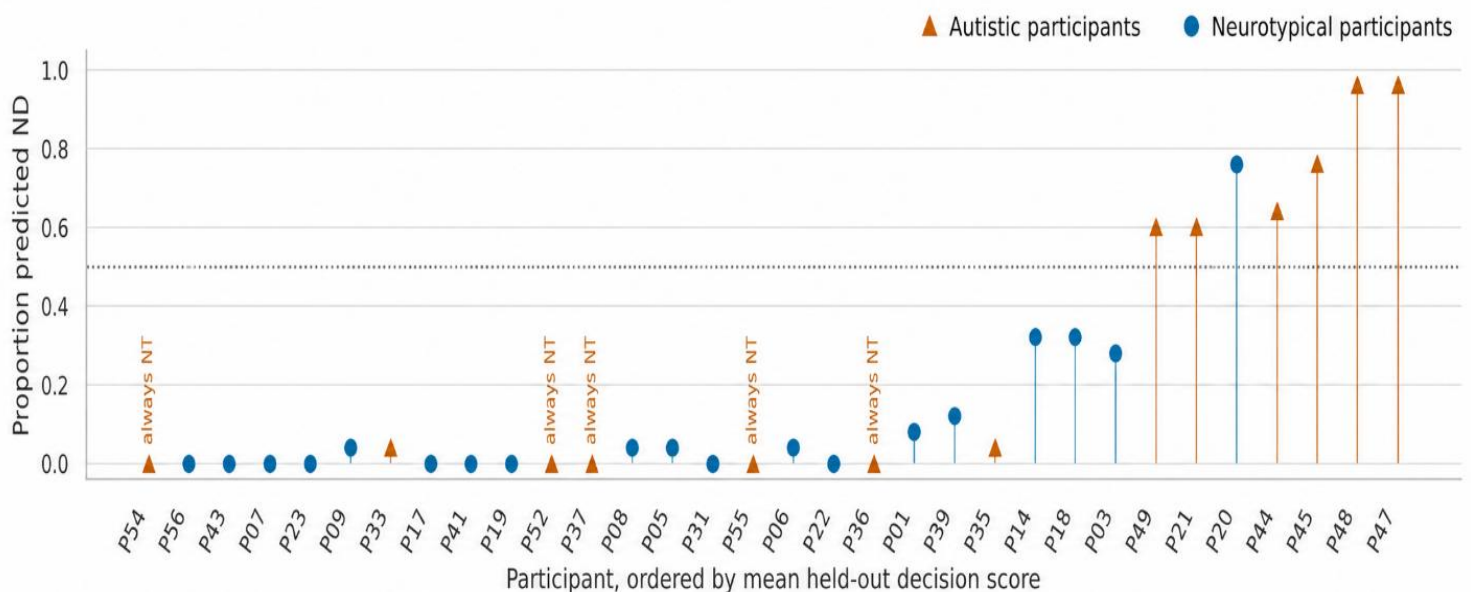
The largest median coefficient magnitude was the dim minus bright response in the right inferior frontal gyrus (-0.92). This coefficient remained negative across all outer fits, with an interquartile range of (-1.16 to -0.78) and a 2.5th-97.5th percentile range of (-1.63 to -0.47). The strongest positive contributions were non-walking minus walking in the right frontopolar cortex ($+0.56$), obstacle minus no-obstacle in the right inferior frontal gyrus ($+0.51$), and non-walking minus walking in the left frontopolar cortex ($+0.45$).

These summaries describe the decision rule repeatedly fitted across different outer-training samples. They do not establish that the corresponding regions differed significantly between groups, nor that large coefficients represent independent biological importance. Because features may be correlated and the selected classifier was not statistically reliable after selection adjustment, the coefficient pattern should be interpreted as an exploratory account of how the model formed its decisions.

5.6 Participant-level prediction stability

Participant-level analysis was used to determine why the strongest observed configuration, full-contrast LDA, achieved high neurotypical specificity but low autistic recall. In particular, the analysis examined whether errors were distributed broadly across the sample or concentrated among specific participants.

Figure 10. Participant-level held-out predictions made by full-contrast LDA. (Here ND refers to autistic participants)



The error pattern was strongly concentrated. Nine participants were classified correctly in all 25 evaluations, and all were neurotypical. In contrast, five autistic participants: P36, P37, P52, P54 and P55 were classified as neurotypical in every evaluation. Two additional autistic participants, P33 and P35, were classified as autistic only once.

The five consistently missed autistic participants contributed 125 of the 261 total errors (47.9%). Including P33 and P35, seven autistic participants accounted for 173 errors (66.3%). Median participant-level accuracy was therefore substantially higher for neurotypical participants at 0.96 (IQR 0.90-1.00) than for autistic participants at 0.04 (IQR 0.00-0.64).

Not all participants were classified consistently. P20, a neurotypical participant, was classified as autistic in 19 of 25 evaluations, while autistic participants P21 and P49 were correctly classified in 15 of 25 evaluations. These cases lay closer to the decision boundary and were more sensitive to changes in the composition of the training data.

Overall, the low autistic recall of full-contrast LDA was not caused by uniformly poor performance across all autistic participants. Instead, it was driven mainly by repeated misclassification of a relatively small subgroup whose fNIRS contrast profiles were consistently placed on the neurotypical side of the decision boundary. The selected model therefore separated some participants reliably, but failed to capture the full variability within the autistic group.

5.7 Exploratory SPQ results

SPQ analysis addresses the three exploratory questions related to SPQ: whether sensory sensitivity is distinguishable between more frequently detected or missed autistic participants, whether the full contrast features are related to SPQ scores, and adding SPQ would improve the classification.

Valid SPQ scores were available for 31 of the 32 participants. One participant with autism who also displayed an absent SPQ score (P45) was excluded from the subgroup and correlation analyses involving the SPQ scores. However, this participant was included in the SPQ-augmented classification analysis with a median imputed SPQ score from the relevant training folds.

Among the 12 autistic participants with valid SPQ data, five were correctly classified in at least half of their repeated held-out predictions, while seven were more frequently misclassified as neurotypical. The median score of 95 (IQR 86-102) was observed in the more frequently detected subgroup versus 103 (IQR 97-118) in the more frequently missed autistic subgroup. Thus, the observed direction suggested greater sensory sensitivity among the more frequently detected subgroup, although the difference was not statistically significant. ($U = 8.5$, $p = 0.166$), and small subgroup sizes limited further interpretation.

Associations between SPQ scores and the 28 contrast-ROI features were examined across the 31 participants with valid scores. The strongest observed Spearman association was with the dim-minus-bright contrast in the left inferior frontal gyrus ($\rho = -0.314$, unadjusted $p = 0.085$). This association did not survive Benjamini-Hochberg correction ($q = 0.881$), and no other SPQ-fNIRS association remained significant after

correction. The analysis therefore provided no evidence that the overall SPQ score was systematically related to any individual contrast-ROI feature.

Figure 11. Summary of metric differences using full-contrast features with and without SPQ.



Adding SPQ produced small, classifier-dependent changes rather than a consistent improvement. LDA performance declined slightly, while linear SVM changed very little. Logistic regression showed the clearest increase in balanced accuracy, ROC-AUC and autistic recall, but neurotypical specificity decreased from 0.807 to 0.767.

Overall, SPQ mainly altered the sensitivity-specificity balance rather than improving separation consistently. Increases in autistic recall were accompanied by lower neurotypical specificity, indicating a redistribution of errors rather than a clear predictive benefit.

5.8 PCA compression and interpretability results

Figure 12 compares the matched full-contrast LDA pipelines with and without PCA. The non-PCA pipeline reproduced the original result exactly, whereas PCA reduced

balanced accuracy from 0.6185 to 0.5881 and ROC-AUC from 0.6965 to 0.6480. MCC decreased from 0.2856 to 0.2175. The reduction was more pronounced for autistic recall, which fell from 0.3475 to 0.2992, than for neurotypical specificity, which decreased from 0.8895 to 0.8770. Across the 100 matched outer fits, balanced accuracy was unchanged in 50 fits, lower with PCA in 35 and higher in 15.

The extent of dimensionality reduction is summarised in **Table 11**. PCA retained a median of five components from the original 28 features, although the selected dimensionality varied substantially across outer fits. Despite this compression, the retained components represented a median of 95.2% of total feature variance.

Figure 12. PCA configuration comparison.

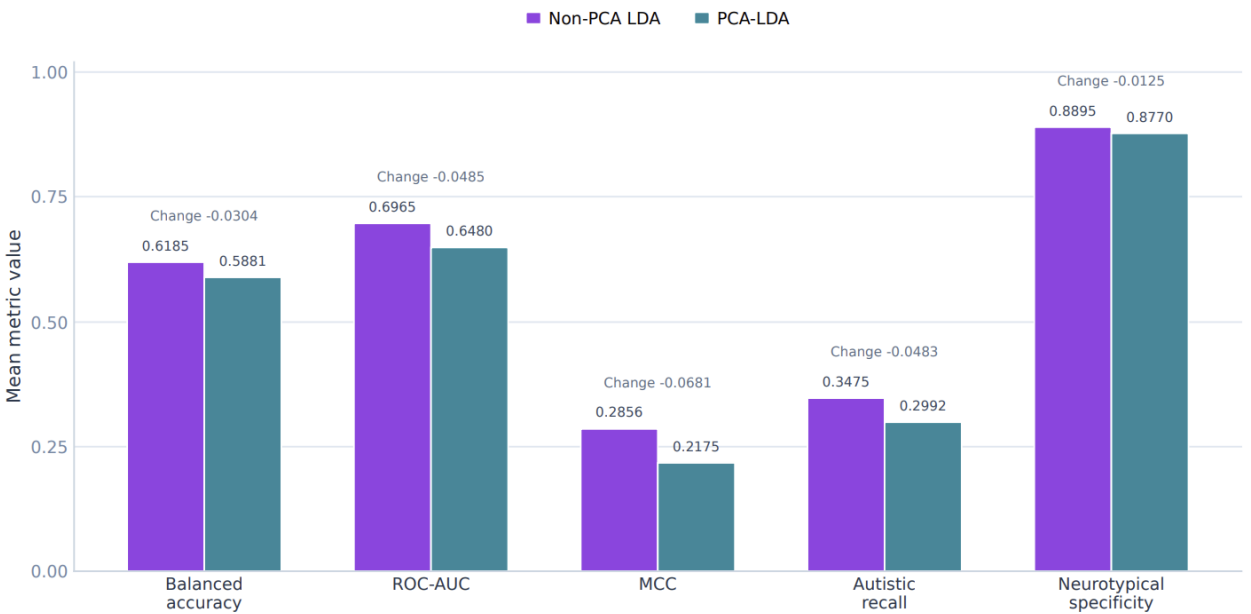


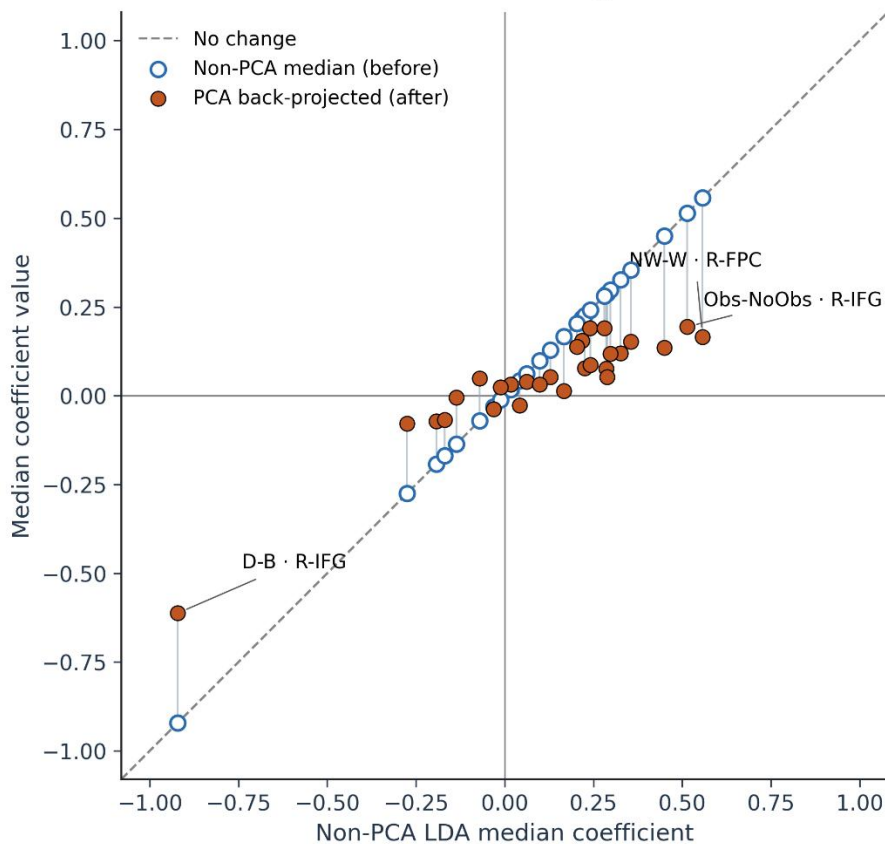
Table 11. PCA component selection and retained variance across 100 outer fits.

Measure	Value
Original feature count	28
Median retained components	5
IQR	3-10
Range	1-12
Mean retained components	6.28
Median retained variance	95.2%
IQR retained variance	93.3 – 98.9%

High variance retention did not correspond to equivalent preservation of the original LDA decision structure. The retained PCA subspace preserved a median of 77.3% of the non-PCA decision direction (IQR 52.9-91.9%). Back-projected PCA and non-PCA coefficient vectors showed a median cosine similarity of 0.7858, Spearman correlation of 0.7031 and sign agreement of 82.1%. The models shared an average of 2.8 of their five highest-ranked features, while feature ranks shifted by approximately six positions. PCA also attenuated the decision function, with a median coefficient-vector norm ratio of 0.5245.

Figure 13. Median LDA coefficients after PCA back-projection.

Median LDA coefficients before and after PCA back-projection
Blue markers show non-PCA values; orange markers show PCA values



DimMinusBright_R_IFG remained the highest-ranked feature in both models and retained the same negative orientation towards the neurotypical side of the fitted boundary. Overall, PCA substantially compressed the representation but preserved predictive performance and the original contrast-ROI decision pattern only partially.

6. Discussion

6.1 Do prefrontal fNIRS features support autistic-neurotypical classification?

The prefrontal fNIRS features provided only limited descriptive support for autistic-neurotypical classification. The strongest observed configuration, full-contrast LDA, achieved balanced accuracy of 0.619, but its performance was markedly asymmetric: neurotypical participants were identified much more consistently than autistic participants. The model therefore did not achieve balanced separation between the two groups. Its apparent performance was driven primarily by reliable recognition of neurotypical participants rather than by dependable identification of autistic participants.

Participant-level prediction stability clarified the source of this asymmetry. Classification was not uniformly poor across all autistic participants. Some were identified consistently, whereas five autistic participants were classified as neurotypical in every held-out evaluation and two others were identified only once across 25 repetitions. Together, these seven participants accounted for 66.3% of all classification errors. The model therefore captured a pattern that applied to a subset of autistic participants but failed to represent the broader variability within the autistic group.

Repeated misclassification should not be interpreted as evidence that these participants were neurologically “more neurotypical.” It indicates only that their contrast-ROI profiles fell on the neurotypical side of the decision boundary learned from the available training participants. Participants whose predictions changed across repetitions were likely positioned closer to an unstable boundary and were consequently more sensitive to changes in training-sample composition. In contrast, consistently missed participants remained within the neurotypical-associated region of the fitted feature space across different training splits. This pattern suggests substantial overlap between the groups together with uncertainty in the boundary estimated from a small sample.

Such overlap is plausible given the heterogeneity of autism. Autistic adults differ considerably in their responses to lighting, movement, colour and visual complexity, making a single uniform prefrontal response unlikely [3]. The source walking study also found that autistic participants reported greater physical and mental challenge despite similar walking duration, illustrating that comparable observable performance can coexist with different subjective or neural demands [15]. However, the present findings indicate that these differences were not expressed consistently enough in the selected ROI beta features to support reliable participant-level separation.

Overall, the observed results were consistent with weak and unevenly distributed group-related structure, but did not establish that such structure was reproducible. Some participants were classified consistently, but low autistic recall, concentrated errors and overlap between participant profiles prevent the fitted pattern from being interpreted as a biomarker or clinically useful classifier. The most defensible conclusion is therefore

that the features showed limited descriptive separability within this sample, without establishing dependable autistic-neurotypical classification.

6.2 Are relative environmental responses more informative than broad task-context averages?

Relative environmental responses were descriptively more informative than broad task-context averages, although this advantage was not statistically established. Because the selection-aware permutation result was non-significant, the following interpretation concerns descriptive differences between fitted feature representations and is intended for hypothesis generation rather than as evidence of reproducible neural group differences. The full-contrast representation produced the strongest observed classification performance, indicating that, within this sample and model search, preserving differences between environmental conditions yielded a more favourable fitted decision rule than broad task-context averaging. This pattern suggests that the classifier made greater use of condition-specific variation than of overall activation averaged across broad periods of walking, but it does not establish that these contrasts capture a reliable autism-related neural signal.

This interpretation is consistent with the usual logic of task-based fNIRS analysis. Within a GLM framework, separate beta coefficients estimate the haemodynamic response associated with each modelled condition, while contrasts are used to examine differences between experimentally meaningful states [9,16]. Most walking-fNIRS studies similarly interpret prefrontal involvement by comparing a simpler walking condition with one that introduces additional cognitive, sensory or motor demands. Pelicioni et al. [5], for example, reviewed 75 comparisons between walking conditions and found that increased task complexity was frequently accompanied by altered prefrontal activation, particularly during secondary verbal, arithmetic or motor tasks. The relevant quantity in these studies is therefore commonly the change produced by additional task demand, rather than activation during walking considered as one uniform condition.

The present experiment differed from conventional dual-task walking studies because complexity was manipulated through illuminance, colour temperature and obstacle presence. Nevertheless, the same principle applies: dim lighting changes the visual information available for navigation, while obstacles increase requirements for detection, foot-placement planning and gait adaptation. The source PEARL analysis also reported selective rather than globally uniform prefrontal effects, including a group-dependent colour-temperature response in the right dorsolateral prefrontal cortex [15]. This suggests that potentially relevant group differences may arise in response to specific environmental manipulations instead of appearing as a stable increase or decrease in overall prefrontal activity. Because the source analysis used the same participants and recordings, it should not be treated as independent confirmation, but it

provides a plausible explanation for why preserving individual condition effects was useful.

The contrast representation also aligns with the possibility that autism-related neural differences are distributed across conditions and regions. Reviews of autism fNIRS research have reported inconsistent directions and locations of activation differences across tasks, rather than one region showing a uniform group effect [10]. The apparent advantage of the contrast features is therefore better interpreted as sensitivity to environmental responsivity than as evidence that one condition or cortical region separated the groups.

This approach was less common than some feature strategies reported in clinical fNIRS machine-learning studies. Eken et al. [11] found that mean HbO amplitudes and HbO-based functional-connectivity measures were among the most frequently used features. The present analysis instead retained GLM-derived condition responses and combined their ROI-level contrasts, preserving a direct relationship between each predictor, the experimental manipulation and its anatomical location. This improved interpretability, but it does not demonstrate that contrast features are generally superior to mean activation, temporal or connectivity features.

Overall, broad task-context averaging may have obscured condition-specific haemodynamic variation that was partly retained by the full contrasts. However, the superiority of one representation over another was not formally tested, and the strongest contrast-based classifier remained non-significant after selection-aware permutation testing. The full-contrast representation should therefore be regarded as the most promising feature formulation within this sample, rather than an established optimal method for autistic-neurotypical classification.

6.3 Did the strongest result exceed random-label expectations after model selection?

The selection-aware result is most informative because it changes the evidential status of the observed classifier. Once several feature representations and classifiers are compared, the relevant benchmark is no longer a nominal balanced accuracy of 0.50. The analysis is effectively asking how well the best result from a model search can perform, and even random labels can yield an apparently promising winner when multiple pipelines are evaluated. The elevated maximum-statistic null distribution therefore shows that moderate above-chance performance can arise from the interaction of small-sample variation and analytical choice, without requiring a stable relationship between the fNIRS features and participant group.

This also implies that the observed full-contrast LDA performance is likely subject to a form of winner's curse. Because it was selected as the highest-performing configuration, its estimated performance is probably more favourable than the performance that would be expected in a new cohort. The issue is not that LDA was fitted incorrectly, but that the selected result benefited from being the most successful of

several noisy estimates. The appropriate interpretation is therefore not that full-contrast LDA narrowly failed to prove a real effect, but that the current data cannot distinguish a reproducible signal from a favourable realisation of the model-search process.

The present result was also lower than many headline values reported in the diagnostic fNIRS machine-learning literature. Eken et al. [11] found that six reviewed studies using nested cross-validation reported accuracies between 73% and 82.5%, with sample sizes ranging from 40 to 363 participants. Studies using leave-one-out cross-validation reported accuracies between 71.72% and 99.30%, while hold-out studies reported values between 65% and 97.6%. The present full-contrast LDA achieved a balanced accuracy of 61.9% and ROC-AUC of 69.7%, placing its descriptive performance below many of these published estimates.

These figures are not directly interchangeable. The reviewed studies involved different clinical populations, tasks, fNIRS systems, feature representations, classifiers and validation procedures. Most reported conventional accuracy, whereas balanced accuracy was used here because the groups were unequal and performance was strongly asymmetric. Ordinary accuracy could obscure low autistic recall by placing greater influence on the more accurately classified neurotypical group. The difference from the published literature therefore cannot be attributed solely to weaker neural information; it may also reflect differences in metrics, study design and the extent to which analytical selection was considered.

Unlike a headline cross-validation score alone, the present analysis also tested whether the selected winner remained unusual after comparison with the best results obtained under randomised labels. Although the observed balanced accuracy exceeded 0.50, it was not unusual relative to the maximum performance produced by the same 12-pipeline search under label randomisation ($p = 0.236$). The main contribution is therefore not high diagnostic accuracy, but a clearer distinction between descriptive model performance and statistically reliable evidence of classification.

6.4 Sensory heterogeneity and the limited value of SPQ

The SPQ analysis did not clearly explain why some autistic participants were detected more consistently by the classifier than others. Although the detected subgroup showed descriptively greater sensory sensitivity, this difference was not statistically reliable. SPQ scores were also not robustly associated with the contrast features, while adding SPQ to the classifiers changed the balance between autistic recall and neurotypical specificity rather than consistently improving overall performance.

This limited contribution may reflect a mismatch between the information captured by the SPQ and the fNIRS task. The SPQ reduces self-reported sensitivity across vision, hearing, touch, taste and smell to one participant-level score, whereas the fNIRS features represented prefrontal responses to specific lighting and obstacle changes during walking. A broad sensory score may therefore fail to reflect how an individual

responds to a particular visual environment while simultaneously navigating and avoiding obstacles.

This interpretation is consistent with Parmar et al. [3], who found substantial variation in autistic adults' visual responses to light, motion, patterns and colours. Participants differed not only in which stimuli affected them, but also in the severity and daily consequences of these experiences. The authors consequently cautioned against generalising sensory findings across the autistic population or assuming that sensitivity in one sensory domain predicts sensitivity in another.

The absence of a clear SPQ benefit therefore does not imply that sensory experience was unrelated to the observed fNIRS patterns. Rather, a single global score was unlikely to capture the context-specific and heterogeneous sensory responses represented by the walking-lighting contrasts. Overall, SPQ provided little additional explanatory or predictive value because its broad summary of sensory sensitivity did not closely correspond to the specific environmental and cognitive demands measured in this study.

6.5 Interpretation of PCA compression and decision-pattern preservation

The PCA analysis was most informative as a test of whether the full-contrast LDA relied on a stable, lower-dimensional structure. Although the retained components represented 95.2% of total feature variance, they preserved only 77.3% of the original LDA decision direction and produced lower balanced accuracy and ROC-AUC. The main implication is therefore that directions of greatest overall variation were not fully aligned with the directions contributing to autistic-neurotypical classification. Because PCA derives its components without using group labels, lower-variance combinations can be removed even when they contribute to the supervised decision boundary [23].

The continued dominance of DimMinusBright_R_IFG showed that one prominent element of the original decision pattern survived compression. However, this feature's preservation was not sufficient to maintain model performance. The decrease in performance despite retention of the highest-ranked feature indicates that the LDA decision was distributed across multiple contrast-ROI relationships rather than being determined by a single regional contribution. PCA produced a recognisably similar overall coefficient direction, but attenuated coefficient magnitude and reordered several secondary features. The result therefore supports partial structural continuity, not preservation of an equivalent decision rule.

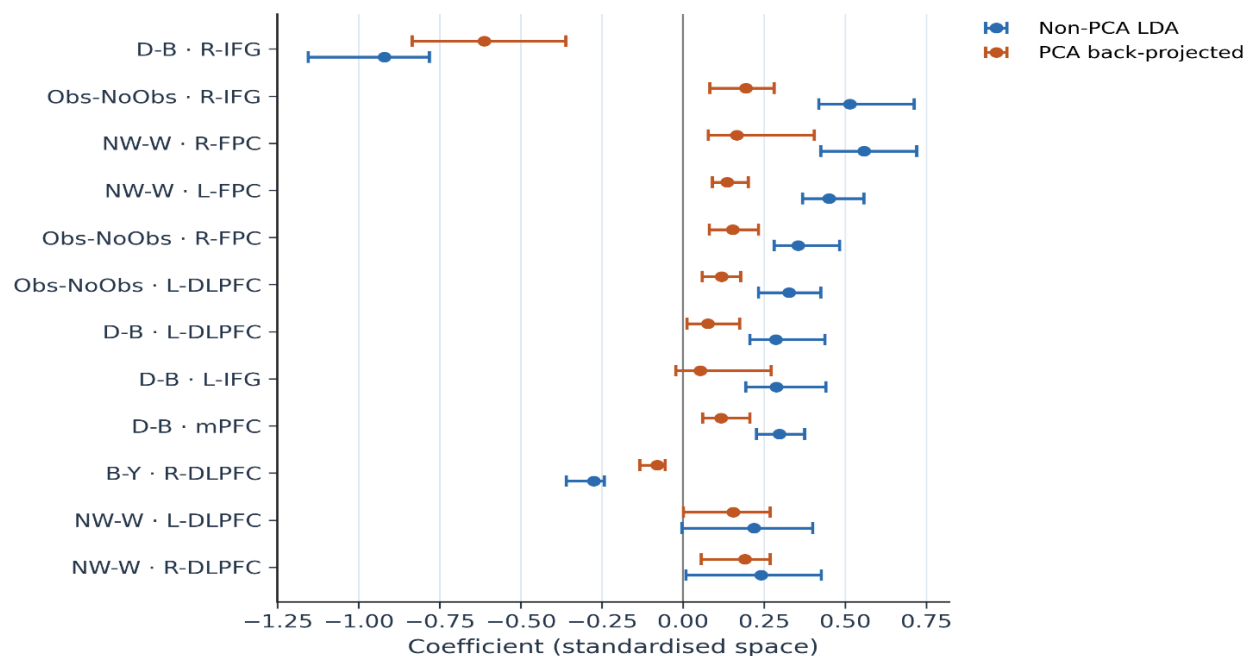
PCA reduced autistic recall more than neurotypical specificity, but the present analysis could not determine which discarded feature combinations caused this change. The result nevertheless shows that PCA did not resolve the classifier's principal failure mode and slightly increased its tendency to classify autistic participants as neurotypical.

Finally, the selected dimensionality varied from one to 12 components across outer fits. This wide variation indicates that the estimated low-dimensional representation was sensitive to the participants included in each training set, rather than revealing a clearly

stable latent structure. Such instability is unsurprising with 32 participants, as small neuroimaging samples produce uncertain cross-validated model estimates even when leakage is controlled. The uncompressed representation was consequently preferable: its 28 features retained explicit experimental and anatomical meaning, while shrinkage LDA already controlled covariance estimation. PCA added an unsupervised information bottleneck without improving performance or stability.

Overall, the PCA analysis showed that the descriptive LDA result was neither reducible to its strongest right-IFG feature nor fully captured by the highest-variance dimensions. Instead, it depended on a weak and distributed multivariate pattern whose structure was only partially robust to compression. Given that the PCA analysis remained exploratory and the primary selection-aware result was non-significant, this pattern should be interpreted as model behaviour rather than evidence of a reproducible neural signature. The broader fNIRS machine-learning literature similarly emphasises the influence of sample size and feature representation on reported classification performance.

Figure 14. Comparison of weighted features in non-PCA and PCA models ordered by non-PCA median absolute rank



6.6 Methodological strengths and limitations

A major strength of this study was the control of information leakage. Standardisation, SPQ imputation, optional PCA and hyperparameter selection were fitted using training participants only, while outer-test participants remained untouched until final evaluation. Repeated nested cross-validation therefore separated model development from performance estimation and reduced the risk of optimistic results caused by tuning and testing on the same data.

Comparisons between pipelines were also designed to be fair and interpretable. The same outer cross-validation splits were used across all classifier-feature configurations, so performance differences were less likely to arise from one model receiving easier test participants. Three regularised linear classifiers were appropriate given the sample size and retained clearer decision rules than the more flexible nonlinear alternatives. Dimensionality was further reduced through ROI aggregation and the use of experimentally defined contrast features while retaining anatomical and task-related meaning.

Other strengths include performance reporting. Balanced accuracy was appropriately chosen as the primary metric in view of the unequal group sizes, but it was interpreted alongside autistic recall, neurotypical specificity, ROC-AUC, and MCC. This prevented a single summary score from obscuring the model's strong neurotypical bias. Repeated held-out predictions and participant-level stability analyses further investigated whether errors were broadly distributed or concentrated among particular participants, thus providing a more informative assessment than average performance alone.

The primary limitation is the small sample of independent participants. Only 32 participants remained, including 13 autistic participants; therefore, each outer-training fold contained approximately 24 participants, whereas some representations had as many as 32 predictors. Although regularisation and nested validation mitigated overfitting, they did not dispel the uncertainty of estimating multivariate decision boundaries from such limited data. Moreover, repeated cross-validation reuses the same participants and, therefore, cannot replace evaluation in an independent cohort. Hyperparameter selection was unstable across outer fits, with no modal value selected in more than 46% of fits. This supports the interpretation in Section 6.3 that the binding constraint was the number of independent participants rather than model complexity.

The available fNIRS representation was thus restricted to prefrontal HbO beta amplitudes. HbR responses, temporal dynamics and functional connectivity were unavailable, so the analysis captured only one aspect of the haemodynamic signal. Averaging 22 channels into seven ROIs improved interpretability and reduced dimensionality, but may also have attenuated informative channel-level variation.

Interpretation of the dim-minus-bright contrast requires additional caution because dim conditions preceded bright conditions in the source experiment. This contrast may therefore partly reflect order, adaptation or fatigue alongside illuminance, which is particularly relevant because DimMinusBright_R_IFG had the largest coefficient magnitude. Finally, although contrast-based representations performed better descriptively, no formal inferential test established that they were superior to the alternative feature sets.

Lastly, class imbalance was handled differently across classifiers. Logistic regression and linear SVM used balanced class weights, whereas LDA estimated empirical class

priors within each training fold. This difference may have contributed to the classifiers' distinct sensitivity-specificity trade-offs and should be considered when comparing their class-specific behaviour.

Overall, the methodology is appropriate, leakage-controlled, and transparent for exploratory small-sample studies. The main limitations are those of the restricted cohort and the inherited feature space, which could not be overcome within the machine-learning pipeline.

6.7 Future work

Future studies should prioritise a substantially larger and more balanced cohort, with enough autistic participants to represent the heterogeneity of sensory and neurophysiological responses. A larger dataset would support more stable model estimation, subgroup analysis and evaluation using an independent external test set. This is particularly important because small fNIRS classification studies can produce unstable or optimistic performance estimates that may not generalise to new participants [11,14].

Future work should also extend beyond regional activation and contrast beta amplitudes. Although these features were interpretable, they did not capture temporal dynamics or communication between prefrontal regions. Previous autism fNIRS research has reported altered functional connectivity, and Chan et al. [24] found reduced lateral-prefrontal connectivity despite similar behavioural accuracy between autistic and typically developing participants. Connectivity-based features may therefore capture neurodevelopmental differences that are less apparent in regional activation magnitude alone [10,24]. Future models could combine activation contrasts with functional connectivity, temporal features and behavioural measures, while retaining nested validation and testing performance in an independent cohort.

7. Conclusion

This dissertation examined whether interpretable prefrontal fNIRS features recorded during environmentally complex walking could support individual-level autistic-neurotypical classification. Four feature representations and three linear classifiers were evaluated using leakage-controlled repeated nested cross-validation and selection-aware permutation testing.

The strongest descriptive result was obtained by LDA using full environmental-contrast features, achieving balanced accuracy of 0.619 and ROC-AUC of 0.697. Across classifiers, contrast-based representations generally outperformed broad task-context averages, suggesting that responses to changes in lighting, obstacle presence and task context may carry more information than overall activation levels. However, no formal inferential test established that contrast features were superior.

Classification performance was strongly asymmetric. Neurotypical participants were identified consistently, whereas autistic recall remained low, with most errors concentrated among a subgroup of autistic participants. The coefficient pattern was distributed across several ROI-contrast relationships rather than being determined by one region or feature. PCA preserved much of the overall feature variance and part of the original decision direction, but reduced performance, indicating that discriminative information was not fully aligned with the highest-variance dimensions. SPQ scores also provided little additional predictive or explanatory value.

Crucially, the strongest classifier did not exceed random-label expectations after accounting for selection across the 12 evaluated pipelines ($p = 0.236$). Non-significance does not demonstrate that the features contain no group-related information; it indicates that such information was not reliably established in this sample. The findings therefore do not support a statistically reliable, generalisable or clinically useful classifier. Instead, the study's main contribution is a transparent and reproducible framework for evaluating interpretable fNIRS features in a small neuroimaging sample. Larger independent cohorts and richer features incorporating HbR, temporal dynamics and connectivity are required to determine whether the observed contrast-based pattern is reproducible.

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Appendix A. Original PEARL walking-lighting experiment and inherited fNIRS processing

Appendix A.1 Study setting and experimental design

The source experiment was conducted at University College London's Person-Environment-Activity Research Laboratory (PEARL), a full-scale facility that enables controlled simulation of pedestrian environments. The study examined walking under systematically varied lighting and environmental conditions while recording walking behaviour and prefrontal haemodynamic activity using functional near-infrared spectroscopy (fNIRS).

The PEARL environment was configured as a controlled urban walking route with adjustable lighting and reconfigurable physical obstacles. Illumination was provided by ETC RAYN Rosa fixtures, which allowed the intensity and spectral composition of the

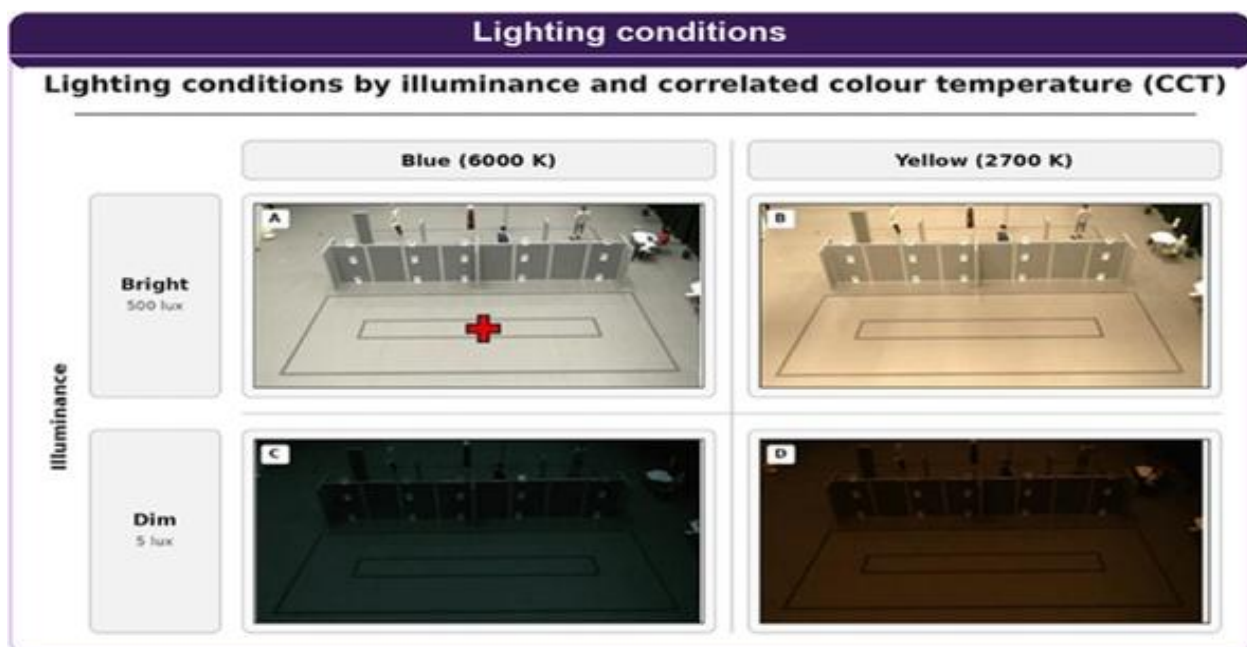
lighting to be manipulated independently while the physical environment remained controlled.

Table A1. Summary of experimental factors.

Factor	Levels	Role in experiment
Illuminance	Bright: 500 lux; Dim: 5 lux	Manipulated lighting intensity
Correlated Colour Temperature	Blue-enriched: 6000 K; Yellow/warm: 2700 K	Manipulated spectral composition
Obstacle presence	Obstacle present; No obstacle	Manipulated environmental complexity

Combining the two illuminance and two correlated colour temperature (CCT) levels produced four lighting conditions: dim-yellow, dim-blue, bright-yellow and bright-blue. The 5-lux conditions represented low-level pedestrian lighting, while the 500-lux conditions provided a substantially brighter comparison. The 2700 K condition produced warmer yellow lighting, whereas the 6000 K condition produced cooler, blue-enriched lighting.

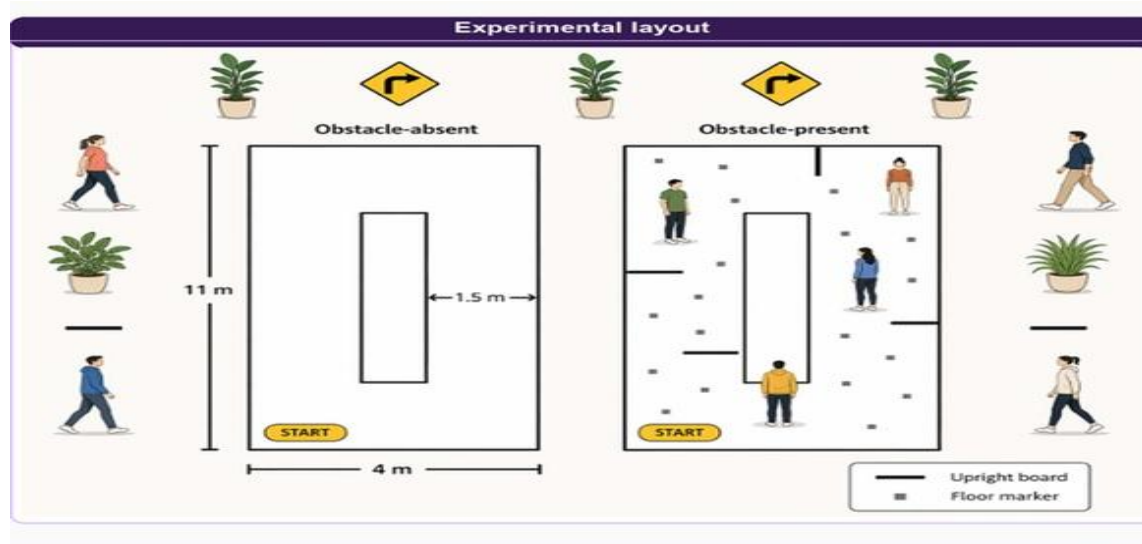
Figure A1. Experimental lighting conditions across illuminance and CCT combinations. Adapted from Davoodian et al. [15].



Participants walked around a rectangular circuit measuring approximately 11 m × 4 m, with a 1.5 m-wide lane marked on the floor. The same route was completed under two configurations. In the no-obstacle condition, participants walked the circuit with minimal

spatial interference. In the obstacle-present condition, the route contained freestanding mannequins, upright boards and small floor-level obstacles. These objects narrowed sections of the route and required participants to adjust their walking path, increasing environmental complexity without changing the overall circuit.

Figure A2. Schematic illustration of walking-path configurations. Adapted from Davoodian et al. [15].



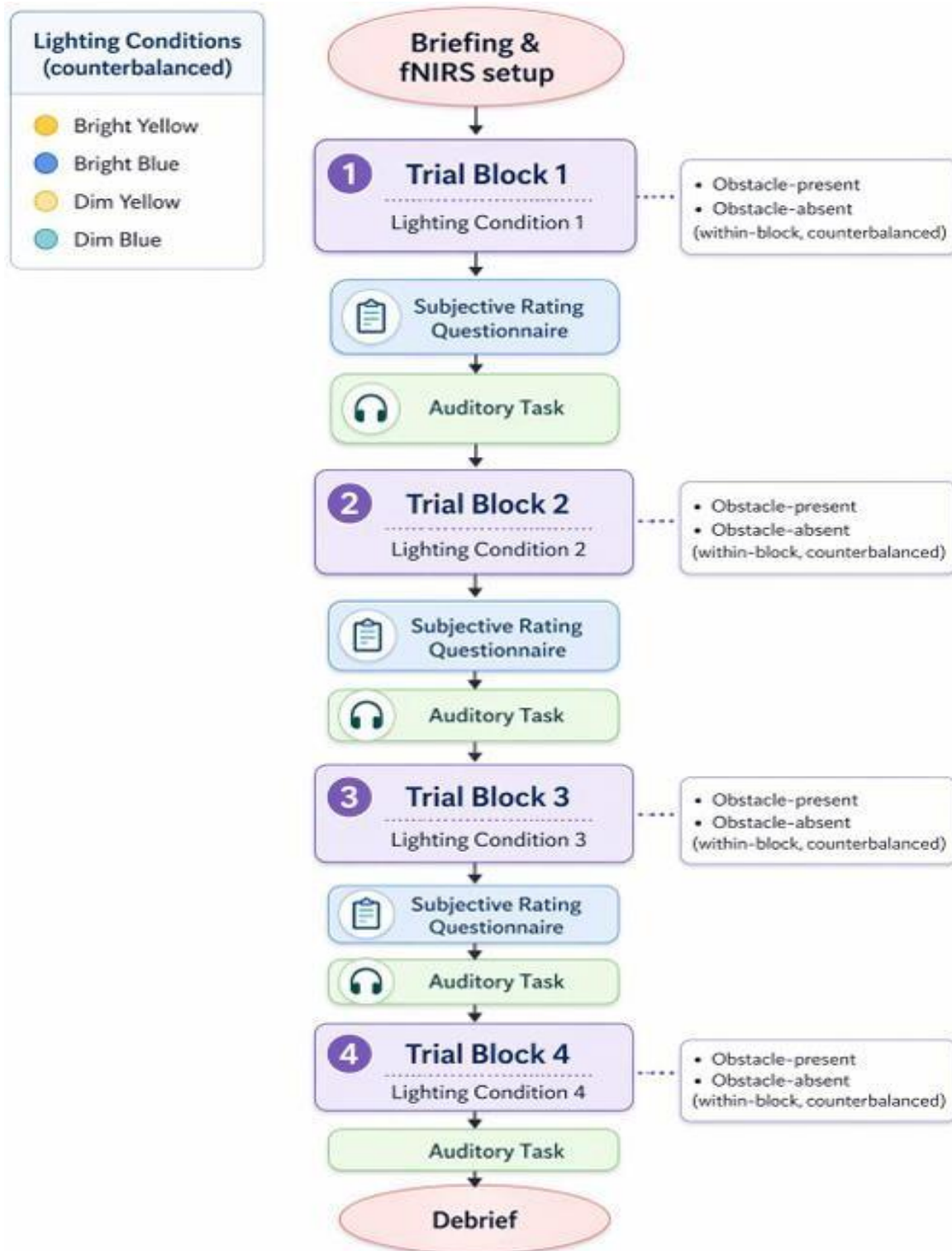
Appendix A.2 Experimental procedure

Participants were fitted with the portable fNIRS cap before completing four lighting-condition blocks. To support visual adaptation, each session began under dim lighting. Participants were exposed to the 5-lux environment for approximately 15 minutes during the experimental briefing and fNIRS cap fitting. The two dim-lighting blocks were completed before the two bright-lighting blocks.

Within each lighting block, participants completed both the obstacle-present and no-obstacle routes. Each trial consisted of four laps completed at a comfortable, self-selected walking pace. Participants were instructed not to rush or deliberately slow down. Walking duration and prefrontal fNIRS activity were recorded continuously throughout each trial.

Approximately six minutes were provided between lighting conditions to allow participants to adapt to the new illumination. During this interval, participants completed a brief auditory attention task. After completing both route configurations within a lighting condition, they rated the comfort of the lighting and the physical and mental challenge of the walking task. The walking trials, auditory periods and transitions were subsequently represented as separate regressors in the first-level GLM.

Figure A3. Experimental task procedure (adapted from Davoodian et al. [15]).



Appendix A.3 Behavioural, subjective and sensory measures

Walking duration was recorded for each trial using a custom experiment implemented in Gorilla.sc. The experimenter manually logged the start and end of each trial using an iPad, and the exported timestamps were used to calculate the time required to complete the four laps.

After completing the obstacle-present and no-obstacle routes under each lighting condition, participants rated:

- the comfort of the lighting;
- the physical challenge of the walking task; and
- the mental challenge of the walking task.

Each response was recorded on a seven-point Likert scale, with higher values indicating greater comfort or greater perceived challenge. These ratings captured participants' immediate experience of each lighting environment alongside the behavioural and fNIRS measurements.

Participants also completed the Sensory Perception Quotient (SPQ), described in Section 4.7.4.

In the present dissertation, the SPQ was examined as a supplementary participant-level measure. Walking duration and subjective ratings provided context for the source experiment but were not included in the primary fNIRS machine-learning feature matrix.

Appendix A.4 fNIRS acquisition details

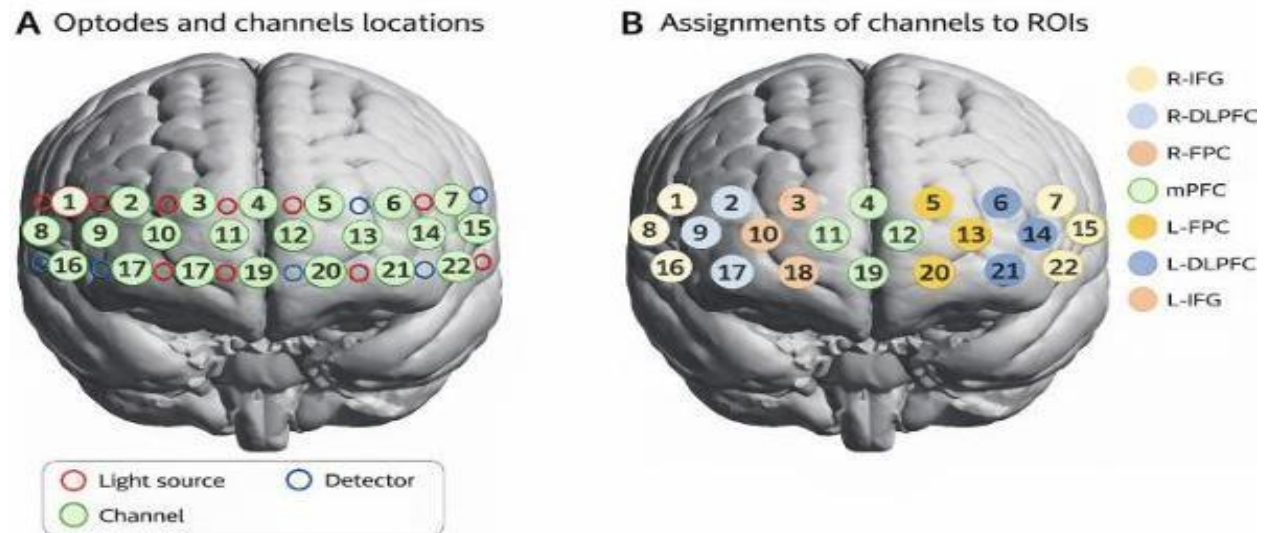
Prefrontal haemodynamic activity was recorded continuously during the walking trials using a portable Shimadzu LIGHTNIRS fNIRS system operated with KNIRS software. The montage comprised eight light sources and eight detectors, forming 22 measurement channels across the prefrontal cortex. Signals were acquired at 13.33 Hz using wavelengths of 780, 805 and 830 nm, enabling estimation of changes in oxygenated haemoglobin (HbO) and deoxygenated haemoglobin (HbR).

Optode placement followed the international 10–20 EEG positioning system. To standardise the montage across participants, Channel 19 was positioned over Fpz at 10% of the nasion–inion distance. The arrangement provided bilateral coverage of the inferior frontal gyrus, dorsolateral prefrontal cortex and frontopolar cortex, together with the medial prefrontal cortex.

Seven regions of interest were defined from the 22-channel layout: right inferior frontal gyrus, right dorsolateral prefrontal cortex, right frontopolar cortex, medial prefrontal cortex, left frontopolar cortex, left dorsolateral prefrontal cortex and left inferior frontal gyrus. In the present dissertation, condition-specific channel beta estimates were

subsequently averaged within these regions to produce anatomically interpretable features.

Figure A4. fNIRS placement on the prefrontal cortex and summary of ROIs. Adapted from Davoodian et al. [15].



Appendix A.5 Signal-quality assessment, first-level modelling and inherited data

In the source experiment, fNIRS preprocessing was conducted in MATLAB using Homer2. Raw intensity signals were visually inspected to identify channels with inadequate signal quality. Channels were excluded when the power spectrum did not contain a clear heartbeat peak around 1.2 Hz, when flat horizontal bands indicated saturation, or when the intensity time series showed severe movement artefacts or near-flat segments consistent with signal loss. These checks reduced the influence of poor optode–scalp contact, saturation and movement-related contamination on the subsequent haemodynamic estimates.

Following quality assessment, the preprocessed HbO time series were downsampled to 1 Hz and modelled in SPM using a channel-wise general linear model. The design matrix included regressors for the eight walking-lighting conditions, auditory attention periods and transitions between experimental blocks. The walking regressors represented the combinations of illuminance, correlated colour temperature and obstacle presence described previously. Each task regressor was convolved with a canonical haemodynamic response function to represent the expected delayed vascular response.

The same design matrix was applied to each of the 22 fNIRS channels, but the GLM was fitted separately to each channel's HbO time series. This produced beta coefficients representing the estimated haemodynamic response amplitude associated with each experimental condition and channel.

The present dissertation did not repeat the raw-signal preprocessing or first-level GLM estimation. Instead, the analysis began from the resulting participant-level SPM.mat files and inherited the condition- and channel-specific HbO beta estimates. These outputs were validated for the required design information, aggregated into seven prefrontal regions of interest and transformed into the task-context and environmental-contrast features evaluated by the machine-learning pipeline.

Appendix B. Mathematical interpretation of inherited GLM beta estimates

For a participant with T fNIRS time points and p regressors, the GLM can be represented as a matrix model in which the measured haemodynamic signal is explained by condition-specific predictors plus residual error.

$$X = \begin{pmatrix} x_1(t_1) & x_2(t_1) & \cdots & x_p(t_1) \\ x_1(t_2) & x_2(t_2) & \cdots & x_p(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(t_n) & x_2(t_n) & \cdots & x_p(t_n) \end{pmatrix}$$

Here, each column x_j is a full time-experiment predictor. For example, one column may represent the expected haemodynamic response to a blue-light walking condition, while another may represent a yellow-light condition, an obstacle condition, an auditory task, a transition period, or a constant baseline term.

For task-based fNIRS, the experimental conditions are commonly first represented as block regressors. A block regressor marks when a condition is active during the experiment. For example, a simplified condition vector $s_j(t)$ may be:

$$s_j(t) = \begin{cases} 1, & \text{if condition } j \text{ is active} \\ 0, & \text{otherwise} \end{cases}$$

However, because the haemodynamic response is delayed and smooth rather than instantaneous, SPM does not usually model the brain response as a simple on/off signal. Instead, the task timing is convolved with haemodynamic response function (HRF). This can be written as:

$$x_j(t) = (s_j * h)(t)$$

Here, the task timing function for each condition is convolved with the haemodynamic response function to produce a smoother expected response shape. The design matrix therefore contains modelled haemodynamic responses rather than raw on/off task labels.

In the source walking-lighting experiment, this modelling approach was applied to auditory attention tasks, walking trials defined by illuminance, colour temperature and obstacle presence, and transition periods between conditions. These events were modelled as block regressors and convolved with a canonical HRF. The GLM was then

fitted separately for each fNIRS channel, and beta values representing task-related haemodynamic changes were estimated for each condition.

The channel-wise nature of the GLM is important for interpreting the resulting SPM-derived values. The same design matrix is shared across channels because the experimental timing is the same for a given participant. However, each fNIRS channel has its own HbO time series. Therefore, a separate beta vector is estimated for each channel:

$$y_i = X\beta_i + e_i$$

For each channel, the HbO time series is modelled using the shared design matrix, a channel-specific vector of beta coefficients and residual error.

For one channel:

$$\begin{bmatrix} y_i(t_1) \\ y_i(t_2) \\ \vdots \\ y_i(t_n) \end{bmatrix} = \begin{bmatrix} x_1(t_1) & x_2(t_1) & \cdots & x_p(t_1) \\ x_1(t_2) & x_2(t_2) & \cdots & x_p(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(t_n) & x_2(t_n) & \cdots & x_p(t_n) \end{bmatrix} \begin{bmatrix} \beta_{1,i} \\ \beta_{2,i} \\ \vdots \\ \beta_{p,i} \end{bmatrix} + \begin{bmatrix} e_i(t_1) \\ e_i(t_2) \\ \vdots \\ e_i(t_n) \end{bmatrix}$$

If the dataset contains 22 fNIRS channels, the same model is repeated for each channel. Therefore, if there are $p = 12$ regressors and $c = 22$ channels, the beta estimates can be arranged as:

$$\beta \in \mathbb{R}^{p \times c}$$

For this project, this can be visualised as:

$$\beta = \begin{bmatrix} \beta_{1,1} & \beta_{1,2} & \cdots & \beta_{1,22} \\ \beta_{2,1} & \beta_{2,2} & \cdots & \beta_{2,22} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{12,1} & \beta_{12,2} & \cdots & \beta_{12,22} \end{bmatrix}$$

Using the approximate dimensions from the SPM files, the full model can be understood as:

$$Y_{6089 \times 22} = X_{6089 \times 12} \beta_{12 \times 22} + \varepsilon_{6089 \times 22}$$

This matrix relationship shows how the same 12 modelled task regressors can be used to explain 22 separate fNIRS channel time series. The beta matrix is therefore a condition-by-channel representation of task-related haemodynamic response.

The beta coefficients are estimated by least-squares fitting. Under ordinary least squares, the estimated beta values are those that minimise the squared difference between the measured signal and the signal predicted by the model:

$$\hat{\beta} = \arg \min ||Y - X\beta||^2$$

For a single channel, assuming X has full column rank, the ordinary least-squares solution is:

$$\hat{\beta}_i = (X^T X)^{-1} X^T y_i$$

For all channels together, this can be written as:

$$\hat{\beta}_i = (X^T X)^{-1} X^T Y$$

More generally, the beta estimate can be expressed using the pseudo-inverse of the design matrix:

$$\hat{\beta} = X^\dagger Y$$

The pseudo-inverse form is useful because neuroimaging design matrices are often not square, and SPM-style implementations store an operator used to map the measured data onto beta estimates. Once the beta values are estimated, residual error can be calculated as the difference between measured and model-predicted signal.

$$\varepsilon = Y - X\hat{\beta}$$

For this dissertation, each beta coefficient can be interpreted as an estimated response amplitude for a specific experimental regressor within a specific fNIRS channel. A positive beta value indicates that the channel's haemodynamic signal varies in the same direction as the modelled task response, while a smaller or negative beta indicates weaker or opposite-direction association, depending on the signal and model context. Importantly, these beta coefficients are not raw HbO values. They are model-derived estimates of condition related haemodynamic response obtained by fitting the task-informed design matrix to each channel's fNIRS time series.

Appendix C. Linear classifier formulations

Let $x_i \in \mathbb{R}^p$ denote the standardised feature vector for participant i , where p was the number of predictors in the relevant feature representation. The class label was coded as $y_i = 0$ for neurotypical participants and $y_i = 1$ for autistic participants. Although all three classifiers produced a linear decision boundary of the general form:

$$f(x) = w^T x + b,$$

They estimated this boundary according to different statistical principles.

Appendix C.1 Linear Discriminant Analysis

Linear discriminant analysis is a generative classification method that models the distribution of the predictors within each class. Following Fisher's discriminant approach, it seeks a linear combination of the input features that separates the class means relative to the variability observed within classes [25]. In the present dissertation, LDA tested whether beta-derived fNIRS feature profiles could be projected onto a decision axis that separated autistic and neurotypical participants.

For class $k \in \{0, 1\}$, LDA assumes that the feature vectors are approximately drawn from a multivariate normal distribution,

$$x | y = k \sim N(\mu_k, \Sigma)$$

Where:

μ_k = Mean feature vector for class k ;

Σ = Covariance matrix shared by both classes;

The model therefore estimates the two class means, one shared within-class covariance matrix, and the class prior probabilities $\pi_k = P(y = k)$.

The discriminant function for class k is:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k).$$

A participant is assigned to the class with the larger discriminant score:

$$\hat{y} = \operatorname{argmax}_k \delta_k(x).$$

$$k \in \{0, 1\}$$

For two classes, the difference between the discriminant functions can be written as one linear decision function,

$$f(x) = w^T x + b,$$

Where:

$$w = \Sigma^{-1}(\mu_1 - \mu_0),$$

And

$$b = -\frac{1}{2}(\mu_1^T \Sigma^{-1} \mu_1 - \mu_0^T \Sigma^{-1} \mu_0) + \log\left(\frac{\pi_1}{\pi_0}\right).$$

Positive decision scores corresponded to classification towards the autistic class. Thus, an LDA coefficient did not represent the effect of a feature independently. It represented the feature's contribution after accounting for the class-mean difference and the covariance between predictors.

Ordinary LDA requires inversion of the estimated covariance matrix. This was problematic for the full-contrast representation because each outer-training set contained approximately 24 participants but included 28 predictors. For a two-class problem, the pooled within-class covariance has a maximum rank of approximately $n_{\text{train}} - 2$. With 24 training participants, its rank could therefore be no greater than approximately 22, making an unregularised 28×28 covariance matrix singular or numerically unstable. The effective fitting sample was smaller still within the inner cross-validation folds.

Shrinkage LDA addressed this problem by replacing the empirical covariance matrix with a weighted combination of the empirical estimate and a scaled identity matrix:

$$\hat{\Sigma}_{\lambda} = (1 - \lambda)\hat{\Sigma}_{sample} + \lambda \left[\frac{tr(\hat{\Sigma}_{sample})}{p} \right] I,$$

Where:

$\lambda \in [0, 1]$ = Shrinkage parameter;

p = number of predictors

$I = p \times p$ identity matrix

When $\lambda = 0$, the estimator reduces to the empirical covariance matrix. As λ increases, off-diagonal covariance estimates are reduced and the variances are drawn towards their overall mean. This improves conditioning and reduces the influence of unstable covariance estimates in small-sample, moderate-dimensional data [26]. Ledoit–Wolf shrinkage is specifically intended to produce a better-conditioned covariance estimate when the number of variables is not small relative to the sample size.

The implementation used the lsqr solver and selected shrinkage within the inner cross-validation loop from

$$\lambda \in \{auto, 0.1, 0.3, 0.5, 0.7, 0.9\}$$

The auto option estimated the shrinkage intensity analytically using the Ledoit–Wolf procedure; scikit-learn supports shrinkage with the lsqr and eigen LDA solvers.

Class priors were not fixed manually. Instead, empirical priors were estimated separately from the class proportions in each training fold:

$$\hat{\pi}_k = \frac{n_k}{n_{train}},$$

where n_k was the number of training participants in class k . No additional class weighting was applied to LDA. Its principal assumptions were therefore approximate class-conditional multivariate normality, a covariance structure shared by autistic and neurotypical participants, and independence between participant observations. Exact normality was unlikely to hold in such a small biological dataset, so LDA was interpreted primarily as a regularised linear classifier rather than as a literal generative description of the population.

Appendix C.2 L2-regularised logistic regression

Logistic regression was used as a second linear classifier [27]. Unlike LDA, it does not model the full distribution of the predictors within each class. Instead, it directly models the conditional probability that a participant belongs to the autistic class.

For participant i , the linear score is:

$$z_i = w^T x_i + b.$$

The logistic, or sigmoid, function transforms this score into an estimated probability:

$$P(y_i = 1 | x_i) = \sigma(z_i) = \frac{1}{1 + \exp(-z_i)}.$$

Equivalently, the model assumes that the log-odds of autistic-class membership are a linear function of the predictors:

$$\log \left[\frac{P(y_i = 1 | x_i)}{1 - P(y_i = 1 | x_i)} \right] = w^T x_i + b.$$

A participant was classified as autistic when the estimated probability was at least 0.5, corresponding to $z_i \geq 0$.

The coefficients were estimated by minimising weighted binary cross-entropy together with an L2 penalty:

$$\text{minimise } \sum_{i=1}^n \alpha_{yi} [-y_i \log(p_i) - (1 - y_i) \log(1 - p_i)] + \frac{1}{2C} ||w||_2^2,$$

Where:

$$p_i = P(y_i = 1 | x_i),$$

α_{yi} = The weight assigned to the participant class

$$||w||_2^2 = \sum_{j=1}^p w_j^2$$

is the squared L2 norm of the coefficient vector.

The L2 term discourages individual coefficients from becoming excessively large. It therefore reduces model variance and limits reliance on unstable feature patterns, while retaining all predictors rather than selecting a subset. The hyperparameter C is the inverse regularisation strength. Smaller values of C impose stronger coefficient shrinkage, whereas larger values allow the model to fit the training observations more closely. The inner loop searched

$$C \in \{0.01, 0.1, 1, 10\}.$$

The implementation used the liblinear solver, L2 regularisation and balanced class weighting. For K classes, balanced weighting assigns class k the weight

$$\alpha_k = \frac{n_{train}}{K n_k}$$

Because $K = 2$, observations from the less frequent class received greater individual weight, while the total weight contributed by each class was approximately equal. Scikit-learn defines balanced weights using this inverse-frequency expression and defines C as the inverse of regularisation strength.

Logistic regression does not require class-conditional normality or equal covariance matrices. Its principal assumptions were that participant observations were independent and that the log-odds of class membership could be represented adequately by a linear combination of the standardised features. The coefficient for one feature was conditional on the other predictors in the model, and substantial correlation between ROI contrasts could therefore affect coefficient magnitude and stability. L2 regularisation reduced, but did not remove, this uncertainty.

Appendix C.3 Linear Support Vector Machine

The third classifier was a support vector machine with a linear kernel. Rather than modelling a probability distribution, the SVM identifies a separating hyperplane with a large margin between the classes [28]. For this formulation, the class labels can be written as $y_i \in \{-1, +1\}$, where $+1$ represents the autistic class.

The decision function is:

$$f(x) = w^T x + b$$

With class prediction:

$$\hat{y} = \text{sign}[f(x)]$$

The soft-margin SVM can be written as the constrained optimisation problem:

$$\text{minimise } (w, b, \xi): \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^n \alpha_{y_i} \xi_i,$$

Subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i,$$

And

$$\xi_i \geq 0.$$

The slack variable ξ_i measures the degree to which participant i lies within the margin or on the incorrect side of the decision boundary. The same formulation can be expressed using hinge loss:

$$\text{minimise } (w, b) \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^n \alpha_{y_i} \max[0, 1 - y_i(w^T x_i + b)]$$

The term: $\frac{1}{2} \|w\|_2^2$

is the margin regularisation term. Because the geometric margin is inversely related to $\| \mathbf{w} \|_2$, minimising this term encourages a wider separating margin. The parameter C controls the trade-off between this wide margin and the penalty assigned to margin violations. A small C places greater emphasis on regularisation and permits more training violations, whereas a large C penalises violations more strongly and may produce a narrower boundary fitted more closely to the training sample.

Support vectors are the training observations with non-zero influence on the fitted boundary. They lie on the margin, inside it, or on the incorrect side of the hyperplane. In terms of the decision function, they generally satisfy

$$y_i f(x_i) \leq 1.$$

Observations located correctly beyond the margin contribute no hinge loss and normally do not affect the final position of the hyperplane.

The linear SVM searched

$$C \in \{0.01, 0.1, 1, 10\}$$

and used balanced class weights. In the SVC formulation, this scales the class-specific penalty according to

$$C_k = C_{\alpha k},$$

so that errors involving the smaller class receive greater weight.

Unlike logistic regression, the SVM decision function does not directly produce class probabilities. In this analysis, `probability=True` was enabled so that continuous autistic-class probability estimates could be used for ROC-AUC calculation. The implementation fitted a sigmoid calibration model, commonly termed Platt scaling:

$$P(y = 1 | f) = \frac{1}{1 + \exp(Af + B)},$$

where A and B were estimated from training data. Scikit-learn's SVC performs this probability-estimation procedure using an internal five-fold cross-validation step. Consequently, the calibrated probability and the uncalibrated SVM class prediction are related but are not mathematically identical and can occasionally be inconsistent.

Because the SVM and its calibration were fitted within the training portion of the nested pipeline, no outer-test observations were used to estimate either the separating boundary or the probability transformation. Nevertheless, the small number of training participants meant that the calibrated outputs should be interpreted as continuous ranking scores for ROC-AUC rather than as clinically validated probability estimates.

The linear SVM did not assume normal predictor distributions or equal class covariance matrices. It also did not require perfect linear separability because slack variables allowed margin violations. It did, however, assume that a useful separation could be

represented by one linear boundary in the standardised feature space and that participant observations were independent.

Appendix D. Software environment and reproducibility information

Table D1 distinguishes the recorded final permutation-run environment from the environment used for the final repository test and document audit.

Table D1. Verified software environment, random seeds and reproducibility information.

Item	Verified value
Python (final permutation run)	3.12.11
Python (final test/audit environment)	3.12.13
scikit-learn	1.8.0
NumPy	2.4.6
pandas	3.0.3
SciPy	1.17.1 in the recorded reproduction environment; 1.18.0 in the final test/audit environment
matplotlib	3.10.9
pytest	9.1.1 in the final test/audit environment
Operating system	Windows 11, 64-bit
Primary and additional seeds	42 for outer CV, inner CV, model initialisation and permutation master seed; permutation-specific seeds were deterministically derived from the master seed
Outer CV	RepeatedStratifiedKFold: 4 folds × 25 repeats = 100 outer evaluations
Inner CV	StratifiedKFold: 3 folds, shuffled, seed 42
Candidate split schedules	Identical deterministic outer schedule across the 12 configurations; final feature tables had identical participant order
Permutation split control	Each permutation used one participant-label vector and deterministic CV seed across all candidate configurations; folds were regenerated for the permuted labels
Permutation workflow	The complete nested workflow, including preprocessing, tuning, fitting and outer evaluation, was rerun for every configuration and permutation
Final permutation configuration	configs/permutation_model_space.json; 1,000 valid participant-label permutations; seed 42

Key scripts	build_spm_features.py (validation/features); run_ml_nested_cv.py (model evaluation); summarise_outer_predictions.py (participant summaries); run_permutation_inference.py (inference); analyse_psq_pipeline.py (SPQ); generate_dissertation_figures.py (verified figures/coefficient summaries)
Tests	7 modules; 87 passed, 0 failed, 2 expected questionnaire-merge warnings; 19.47 s
Item	Verified value
Repository provenance	Commit c7f1b8a0bf0afe8b7e9c4edec99bc3abe31c7091 on feature/spm-preprocessing; working tree contained documented uncommitted changes
Recorded reproduction check	The reproduced full-contrast LDA summary, predictions, fold metrics and best parameters matched the saved originals by SHA-256

Appendix E. Declaration and acknowledgement of generative artificial intelligence use

Generative artificial intelligence tools, including **OpenAI ChatGPT and Codex** and **Anthropic Claude**, were used as assistive tools during the preparation of this dissertation. Their use was limited to supporting the presentation, verification and efficiency of work undertaken by the author.

AI assistance was used for the following purposes:

- Producing and refining Python code used to **generate figures** from the author's analysis outputs;
- summarising author-generated CSV files into draft metric tables and written descriptions for results reporting;
- refactoring analysis and utility scripts to reduce duplication, improve readability and streamline computational efficiency;
- proofreading for grammar, clarity, consistency, figure numbering and terminology;
- assisting with the placement and formatting of in-text citations and checking consistency between citations and the reference list.

All numerical results reported in the dissertation were derived from the project data and analysis pipeline. AI-generated or AI-modified code was inspected, executed and tested by the author, and the resulting values were checked against the underlying CSV files and analysis outputs. Figures produced with AI assistance were based on the author's data and specified analyses rather than synthetically generated scientific results. References and in-text citations suggested or formatted with AI assistance were checked against the corresponding original publications before inclusion.

Generative AI was **not** used to collect the experimental data, assign participant group labels, determine participant inclusion or exclusion, perform the original fNIRS preprocessing or first-level GLM estimation, independently select the final statistical methodology, or make unreviewed decisions about the interpretation of the findings. It did not replace examination of the primary literature, critical evaluation of the results, or consultation with the project supervisors. The research questions, analytical decisions, interpretation of the findings and conclusions presented in this dissertation remain the work and responsibility of the author.

The author recognises that generative AI systems can produce inaccurate statements, incorrect citations, inappropriate statistical interpretations and plausible but faulty code. AI outputs were therefore treated as suggestions rather than authoritative evidence. No output was accepted solely on the basis that it had been generated by an AI system, and final responsibility for the accuracy, integrity and academic content of the dissertation rests with the author.

Appendix F. Code repository

The analysis code, aggregate outputs and reproducibility documentation are available in a public GitHub repository at:

<https://github.com/LyCoda/fnirs-walking-autism-classification>